

72nd BPSC PRELIMS

प्रहार

Test Series

Test-19/3

FULL LENGTH TEST

Solution

English



BPSC TEST-19/3 (SOLUTION)

Q 1. Ans. B

Explanation:

- **Geographical Location:** Situated on the fertile Kachi (Kacchi) plain of Balochistan, Pakistan, located near the strategically important Bolan Pass—a traditional overland gateway connecting the Indus Valley with the Iranian Plateau and West Asia.
- **Chronological Signposts:** Spans from circa 7000 BCE to 2500 BCE. It bridges the Aceramic Neolithic (pottery-less farming) phase, the Ceramic Neolithic phase, and the Chalcolithic transition leading up to the Indus Valley Civilisation.
- **Subsistence & Agrarian Economy:** Provides the earliest recorded evidence of agriculture and animal pastoralism in the South Asian subcontinent. Inhabitants cultivated six-row barley, einkorn wheat, and emmer wheat, and transitioned from hunting wild animals to domesticating zebu cattle, sheep, and goats.
- **Settlement & Architecture:** Features well-planned, permanent settlements consisting of rectangular or square houses built from sun-dried mud bricks. Houses contained multiple internal compartments, which were systematically utilised as granaries to store agricultural surplus.
- **Prehistoric Technology & Crafts:** Holds the earliest evidence of cotton cultivation and spinning in the ancient world. Craft specialisation is marked by the production of standard terracotta figurines, glazed faience beads, and extensive lithic blade industries using flint.
- **Burial Practices & Beliefs:** Human burials show skeletons interred in a flexed position, frequently accompanied by elaborate grave goods such as bitumen-lined baskets, personal ornaments made of imported lapis lazuli, turquoise, sea-shells, and sacrificed young goats, indicating a structured belief in the afterlife.

Q 2. Ans. C

Explanation:

- **Mehrgarh:** Located in the Kachi plain of Balochistan, this site yields the earliest archaeological proof of a settled agrarian lifestyle in the Indian subcontinent, with foundational levels dating back to 7000 BCE.
- **Crops cultivated:** The earliest inhabitants systematically grew wheat and barley. This

stable agrarian base eventually supported the demographic expansion leading to the urbanised Harappan civilisation.

- **Animal domestication:** Alongside farming, the people of Mehrgarh domesticated sheep, goats, and cattle. This marks a definitive shift from a nomadic hunting and gathering economy to a settled, food-producing society.
- **Koldihwa:** While Mehrgarh represents the earliest evidence for wheat and barley, Koldihwa in the Belan Valley of Uttar Pradesh is historically significant for yielding some of the earliest evidence of domesticated rice cultivation, dated approximately to 6000 BCE.
- **Burzahom:** Located in the Kashmir Valley, this Neolithic site is distinct for its underground pit dwellings and the unique ritualistic practice of burying domestic dogs alongside human graves. It belongs to a later chronological bracket than Mehrgarh.
- **Chirand:** Situated in the alluvial plains of Bihar, this site is renowned for a massive assemblage of specialised bone tools crafted primarily from deer antlers.

Q 3. Ans. C

Explanation:

- **Northern Black Polished Ware (NBPW)** dates between the 6th century BCE and the 2nd century BCE, perfectly coinciding with the Second Urbanisation and the rise of the Mahajanapadas.
- **Features:** It is a glossy, metallic-looking ceramic made from fine clay on a fast wheel. The distinct mirror-like black finish was achieved by firing the pottery under intense, reduced conditions in a closed kiln.
- The NBPW phase marks a period of profound economic change, running parallel to the widespread agricultural use of iron ploughshares, the introduction of punch-marked coins, and the growth of trade guilds.
- **Luxury items:** Archaeologists categorise NBPW as elite tableware (primarily bowls and small dishes). Excavated shards found repaired with copper pins prove it was highly valued and too expensive to simply discard.
- **Geographical spread:** Originating in the Middle Gangetic valley (sites like Pataliputra and

Kaushambi), it spread as far as Taxila in the north-west and Amaravati in the south, reflecting the vast trade networks of the Mauryan Empire.

- In Indian archaeology, the NBPW phase directly succeeds the Painted Grey Ware (PGW) culture, the latter being associated with the preceding Later Vedic period.

Q 4. Ans. A

Explanation:

- The Harappan civilisation employed a highly standardised system of weights to regulate commercial exchanges.
- **Progression system:** The lower denominations of weights followed a strict **binary progression** (1, 2, 4, 8, 16, 32, up to 12,800).
- The higher denominations shifted to a decimal system, operating primarily on multiples of 10 and 16.
- **Material and shape:** Weights were predominantly manufactured from a hard, flint-like stone called chert. They were strictly cubical in shape and devoid of any numerical markings or inscriptions.
- **Standardisation:** The 16th ratio, weighing approximately 13.65 grams, formed the most common foundational unit of exchange. This structural uniformity was stringently enforced across distant urban centres like Mohenjodaro, Harappa, and Lothal.
- **Application:** Smaller weights were exclusively utilised for measuring high-value, low-volume commodities such as gold, beads, and semi-precious stones. Metal scale pans recovered from numerous excavations substantiate the high precision of their trading networks.
- **Sexagesimal progression:** Archaeological evidence confirms extensive maritime trade between the Harappans (referred to as Meluhha) and Mesopotamia. Mesopotamia utilised a sexagesimal (base-60) system. However, Harappans strictly maintained their indigenous binary-decimal system.
- **Rati seed progression:** Rati (gunja berry) seed formed the foundational weight unit for later Early Historic Indian metrology, particularly in the minting of punch-marked coins (karshapanas). However, Harappan weights were standardised chert cubes using the 16th ratio (approx 13.65 grams) as the base, completely predating the ratio-based system.

Q 5. Ans. C

Explanation:

- **Pair 1 is correctly matched: Gavishti → A literal search or war fought for cows**
 - ✓ In the Rig Veda, the term Gavishti literally means “search for cows” and came to denote conflicts or wars undertaken for the acquisition of cattle, reflecting the pastoral nature of the Rig Vedic economy where cattle constituted the principal form of wealth.
- **Pair 2 is correctly matched: Dasyus → Indigenous non-Aryans who did not perform Vedic sacrifices and spoke a different tongue**
 - ✓ The Dasyus are described in the Rig Veda as people who did not perform Vedic rituals (akratu), were non-sacrificing (ayajvan), and followed customs different from those of the Vedic Aryans. They are often interpreted as indigenous groups or communities outside the Vedic cultural fold.
- **Pair 3 is correctly matched: Sabhapati → The specialised presiding officer of the Rig Vedic tribal assembly (Sabha)**
 - ✓ The Sabha was an important tribal assembly of the Rig Vedic period, and the Sabhapati was its presiding officer. Literary evidence suggests that women could participate in the Sabha, indicating a relatively flexible social structure in the early Vedic age.

Q 6. Ans. D

Explanation:

- **Pair 1 is correctly matched: Kshetra → Cultivable land ready for agriculture**
 - ✓ In Gupta-period land records, Kshetra referred to arable land already under cultivation or fit for immediate agricultural use.
- **Pair 2 is correctly matched: Khila → Waste or uncultivable fallow land**
 - ✓ Khila denoted fallow, barren, or uncultivated land that was not being used for regular agricultural purposes.
- **Pair 3 is correctly matched: Aprahata → Dense, uncultivated forest or wild land**
 - ✓ The term Aprahata literally means “unploughed” or “untouched” land and refers to forest tracts or virgin land not yet brought under cultivation.

Q 7. Ans. B

Explanation:

- The Rahnumai Mazdayasnan Sabha was established

in 1851 in Bombay by reformers such as Dadabhai Naoroji, Naoroji Furdonji, and S.S. Bengalee. Its objective was to reform and modernise Parsi society by promoting education, especially women's education, discouraging social evils, and encouraging a rational interpretation of Zoroastrian religious texts.

- The Sabha played a role similar to that of other 19th-century socio-religious reform movements, seeking to reconcile traditional beliefs with modern ideas.

Q 8. Ans. C

Explanation:

- The Vernacular Press Act of 1878, enacted during Lord Lytton's viceroyalty, was nicknamed the "Gagging Act" due to its highly discriminatory and draconian provisions targeted exclusively at non-English language newspapers.
 - ✓ **Statement 1 is correct:** The District Magistrate was empowered to call upon the printer and publisher of any vernacular newspaper to enter into a bond. The magistrate could demand a security deposit, which could be confiscated if the newspaper published material deemed seditious or likely to stir up disaffection against the government.
 - ✓ **Statement 2 is incorrect:** The Act intentionally bypassed judicial oversight. The decision of the District Magistrate was considered absolute and final, and no appeal could be made in a court of law against it.
 - ✓ **Statement 3 is correct:** As per the structural framework of the Act, a provision existed where a vernacular newspaper could obtain an exemption from entering into the restrictive security bond. To get this exemption, the publisher had to agree to submit the proofs of the newspaper to a government censor before publication.

Q 9. Ans. B

Explanation:

- **Tattwabodhini Sabha (1839):** Founded by Debendranath Tagore (father of Rabindranath Tagore) in Calcutta. Before joining the Brahmo Samaj, he established the Tattwabodhini Sabha to promote a systematic study of India's past and propagate Rammohan Roy's ideas based on the Upanishads.
- **Faraizi Movement (1818):** Founded by Haji Shariatullah in 1818 in Eastern Bengal. It was an Islamic revivalist movement that aimed at purging un-Islamic practices and restoring the pristine purity of Islam. Under Shariatullah's son, Dudu

Miyan, the movement also took on a revolutionary, socio-economic character against oppressive zamindars and British indigo planters.

- **Prarthana Samaj (1867):** Established in Bombay by Dr. Atmaram Pandurang, with prominent later leaders like Mahadev Govind Ranade and R.G. Bhandarkar. It was closely allied with the Brahmo Samaj and focused on social reforms like inter-caste dining, widow remarriage, and the upliftment of women and depressed classes.
- **Satyashodhak Samaj (1873):** Founded by Jyotiba Phule in Pune, Maharashtra. The Truth-Seekers' Society was aimed at liberating the weaker sections of society, particularly the Shudras and Ati-Shudras, from priestly exploitation and Brahminical dominance.

Q 10. Ans. D

Explanation:

- Before the establishment of the Indian National Congress in 1885, various political associations such as the Indian Association, the Poona Sarvajanik Sabha, and the Bombay Presidency Association organised public campaigns against several colonial policies.
 - ✓ **Imposition of import duty on cotton:** In 1875, a major public agitation was launched against the British government's decision to appease Manchester textile manufacturers by removing import duties on British cotton goods. This step severely harmed the nascent Indian textile industry, leading to widespread protests demanding protectionist fiscal policies for Indian industries.
 - ✓ **Indianisation of government service:** Throughout the 1870s and 1880s, nationalists organised sustained campaigns for the "Indianisation" of civil services. The issue escalated into a major agitation in 1877 when Lord Lytton lowered the maximum age limit for the Civil Service Examination from 21 to 19 years, an intentional barrier to keep educated Indians out of the administration. Surendranath Banerjee and the Indian Association led an all-India protest against this move.
 - ✓ **Lytton's Afghan adventure:** Lord Lytton's aggressive, expansionist foreign policy led to the Second Anglo-Afghan War (1878–1880). Indian nationalist leaders heavily criticised and agitated against this "Afghan adventure," pointing out that the enormous financial burden of the war was being funded entirely by impoverished Indian

taxpayers while millions were simultaneously dying in severe domestic famines.

- ✓ **Arms Act (1878):** The Indian Arms Act of 1878 made it a criminal offence for Indians to carry or possess arms without a license, whereas Europeans were completely exempted from this rule. This flagrant racial discrimination sparked fierce political agitation and public condemnation across the country.

Q 11. Ans. B

Explanation:

- In 1878, Viceroy Lord Lytton lowered the maximum age limit for candidates appearing in the Indian Civil Service (ICS) examination from 21 years to 19 years.
- The policy was designed to restrict the entry of educated Indians into the covenanted civil services. Competing in London, mastering classical European languages, and bearing high travel costs were already systemic barriers. The lowered age limit compounded these hurdles, making qualifying exceptionally difficult for Indian candidates.
- Surendranath Banerjee and the Indian Association organised an all-India campaign against the age reduction, leading to early pan-Indian political mobilisation prior to the formation of the Indian National Congress.
- Parallel to this reduction, Lytton introduced the Statutory Civil Service in 1879. It reserved one-sixth of covenanted posts for Indians of aristocratic descent through nomination rather than open competition, though the scheme lacked credibility and eventually failed.

Q 12. Ans. B

Explanation:

- Shahu, the son of Sambhaji and grandson of Shivaji, was captured by Mughal forces in 1689 along with his mother, Yesubai, following the execution of his father. He remained a Mughal prisoner for 18 years under the reign of Aurangzeb.
- Following Aurangzeb's death in 1707, Shahu was released. While it was initially Prince Azam Shah who permitted his return to the Deccan on the advice of Mughal noble Zulfiqar Khan, Bahadur Shah I confirmed his release and status after securing the Mughal throne.
- Bahadur Shah I granted Shahu the right to collect sardeshmukhi in the Deccan but deliberately withheld the right to collect chauth, a move designed to keep the Maratha leadership dependent and unsettled.

- The release was a calculated strategic measure intended to ignite a civil war within the Maratha camp. It successfully created a succession struggle between Shahu and his aunt Tarabai, who had been administering the Maratha state on behalf of her minor son, Shivaji II.
- The internal division culminated in the Battle of Khed (1707), where Shahu defeated Tarabai's forces, paving the way for him to crown himself at Satara and establish the Peshwa period of Maratha history.

Q 13. Ans. C

Explanation:

- In 1755, Haidar Ali was appointed the Faujdar of Dindigul by the Mysore administration. During his tenure there, he recognised the importance of modern military technology and, with the assistance of French experts, established a modern arsenal at Dindigul.
 - ✓ The arsenal manufactured and maintained firearms, ammunition, and military equipment on European lines.
 - ✓ This experience significantly contributed to Haidar Ali's subsequent military reforms and rise to power in Mysore.
 - ✓ The adoption of French military techniques later enabled Mysore to emerge as one of the strongest Indian powers resisting British expansion in South India.

Q 14. Ans. A

Explanation:

- Saadatullah Khan served as the Nawab of the Carnatic. Before his death in 1732, he adopted his nephew, Dost Ali Khan, and nominated him as his successor to the throne.
- The Carnatic was officially a subordinate province under the Deccan Viceroyalty, which was governed by Nizam-ul-Mulk Asaf Jah I of Hyderabad. Saadatullah Khan executed this succession completely without the required permission or formal confirmation from the Nizam.
- This unilateral nomination was a significant violation of the Mughal administrative framework. It effectively transformed the governorship of the Carnatic from an appointed imperial office into an independent, hereditary possession.
- This act of defiance deeply angered the Nizam and fractured regional stability. The resulting dynastic disputes later paved the way for a complex war of succession between rival claimants like Chanda Sahib and Anwar-ud-Din.

- The internal political vacuum created by these independent succession struggles provided the British and the French East India Companies the perfect opportunity to intervene politically and militarily, culminating in the Carnatic Wars.

Q 15. Ans. D

Explanation:

- The slogan “No taxation without representation”, originally associated with the American War of Independence, was adopted by the early Indian nationalists in their demand for greater financial control and representative institutions.
- The Moderates argued that Indians were being subjected to heavy taxation without having an adequate voice in the legislative bodies that determined fiscal policies.
- They demanded elected representation in legislative councils, greater control over public expenditure, and the right to scrutinise the colonial budget.

Q 16. Ans. B

Explanation:

- The First Anglo-Sikh War (1845 to 1846) formally concluded with the signing of the Treaty of Lahore on 9 March 1846.
- The Sikh Empire was forced to cede the strategically and economically vital Jalandhar Doab (the territory situated between the Beas and Sutlej rivers) to the British East India Company.
- The British imposed a massive war indemnity of one and a half crore rupees. Unable to pay the full sum, the Lahore Durbar ceded the provinces of Kashmir and Hazara, which the British subsequently sold to the Dogra noble Gulab Singh for 75 lakh rupees under a separate agreement.
- The treaty systematically dismantled Sikh military power by severely limiting the maximum size of the Khalsa army and confiscating all field artillery used against the British during the war.
- A British Resident, Sir Henry Lawrence, was installed at the Lahore Durbar. While the minor Duleep Singh was recognised as the Maharaja under the regency of Rani Jindan, the British effectively established paramountcy over the internal and external affairs of the state.
- Dissatisfaction with these terms soon led to the supplementary Treaty of Bhairawal in December 1846, which completely sidelined Rani Jindan and placed the administration under a Council of Regency controlled entirely by the British Resident.

Q 17. Ans. C

Explanation:

- In 1814, the British railway engineer George Stephenson constructed a steam-powered rail locomotive named the ‘Blucher’.
- It was specifically built to haul heavy loads of coal from the mines to the ports. The engine was capable of pulling a weight of 30 tons up a hill at a speed of 4 miles per hour.
- **Stephenson successfully combined two critical innovations:** mechanical haulage by a steam engine and the use of iron tracks, which had started replacing wooden tracks in the 1760s.
- This 1814 design laid the foundation for the transport revolution. It led directly to the inauguration of the first public railway line connecting the cities of Stockton and Darlington in 1825.
- While Richard Trevithick had earlier built the Puffing Devil in 1801, the world’s first steam-powered road locomotive, which carried passengers on a road in Camborne, Cornwall, and his 1804 Penydarren locomotive was the world’s first steam locomotive to run on rails, these designs were not commercially viable.
- James Watt is credited with significantly improving the stationary steam engine in 1769, but he did not design railway locomotives.
- Isambard Kingdom Brunel was a later civil engineer known for the Great Western Railway and steamships.

Q 18. Ans. D

Explanation:

- Under the Permanent Settlement introduced by Lord Cornwallis in 1793, the total land revenue collected from the cultivators was strictly divided into eleven parts.
- The British East India Company claimed ten-elevenths of the total collected rental as the fixed revenue share for the state.
- The zamindars were officially permitted to retain the remaining one-eleventh of the rental. This served as their commission for carrying out the administrative task of revenue collection.
- In exchange for guaranteeing this fixed revenue stream, zamindars were elevated from ordinary revenue collectors to recognised proprietors of the soil, gaining hereditary and transferable ownership rights over their estates.
- The system operated under strict penal conditions. If a zamindar failed to deposit the Company’s exact

ten-elevenths share before sunset on the prescribed date, their land rights were confiscated, and the estate was auctioned to the highest bidder to recover the arrears.

- Since the initial state demand was fixed at an exceptionally high rate, many zamindars frequently failed to collect the required amount from the peasantry. This led to massive defaults and the widespread auctioning of traditional zamindari estates in the early decades of the settlement.

Q 19. Ans. C

Explanation:

- **Statement 1 is correct:** Lord Cornwallis introduced significant police reforms and relieved the zamindars of their traditional police functions. The responsibility for maintaining law and order was transferred from local intermediaries to a more organised state-controlled police system.
- **Statement 2 is incorrect:** Cornwallis divided districts into circles or thanas, each placed under a Daroga (police officer). These thanas were supervised by the District Magistrate, not by a District Superintendent of Police. The post of District Superintendent of Police emerged much later under British administration.
- **Statement 3 is correct:** Under the Cornwallis system, Indians were excluded from superior posts in the police and other branches of administration. Higher positions were largely reserved for Europeans, while Indians were confined to subordinate roles.

Q 20. Ans. B

Explanation:

- **Statement 1 is incorrect:** Under the dyarchy system introduced by the Government of India Act 1919, provincial subjects were divided into two categories. Education, public health, and local self-government were classified as 'Transferred' subjects, administered by the Governor on the advice of Indian ministers responsible to the legislature. The 'Reserved' subjects (like police, justice, and land revenue) were kept under the Governor and his Executive Council.
- **Statement 2 is correct:** The Act introduced a bicameral legislature at the centre. The lower house, the Legislative Assembly, was constituted with a total strength of 145 members, out of which 104 were elected representatives, ensuring a clear elected majority.
- **Statement 3 is correct:** Addressing a long-standing

demand of Indian nationalists regarding the drain of wealth, the Act mandated that the salaries of the Secretary of State for India and his establishment would henceforth be paid out of British revenues instead of Indian revenues.

- **Statement 4 is correct:** The upper house, the Council of States, had a maximum sanctioned strength of 60 members. Of these, 34 were strictly elected, and 26 were nominated, maintaining an elected majority in the upper chamber as well.

Q 21. Ans. B

Explanation:

- The New English School was established in 1880 at Poona (Pune) by a group of nationalist educationists, including Bal Gangadhar Tilak, Gopal Ganesh Agarkar, Vishnushastri Chiplunkar, and Mahadev Ballal Namjoshi. The institution aimed to provide quality education rooted in national values and free from excessive government control.
- To expand their educational efforts, these reformers founded the Deccan Education Society in 1884. Under its aegis, the New English School later developed into Fergusson College in 1885, which became one of the leading centres of higher education in western India and played an important role in nurturing nationalist thought.

Q 22. Ans. B

Explanation:

- **Observance of the first Khilafat Day (17 October 1919):** Following the end of World War I, Indian Muslims organised a mass agitation against the dismantling of the Ottoman Empire. The All India Khilafat Committee called for a nationwide strike, and 17 October 1919 was observed as Khilafat Day with fasting, prayers, and public meetings.
- **The joint Hindu-Muslim conference (23-24 November 1919):** The All India Khilafat Conference convened a joint session of Hindus and Muslims in Delhi. Mahatma Gandhi was elected as the president of this conference. It was during this session that Gandhi first proposed the strategy of non-cooperation against the British government if the Khilafat demands were not met.
- **The Nagpur session of the Congress (December 1920):** The annual session of the Indian National Congress held at Nagpur definitively endorsed the Non-Cooperation programme. This session ratified the earlier resolution passed at the special Calcutta session in September 1920 and

reorganised the Congress constitution to facilitate mass mobilisation.

- **The Chauri Chaura incident (4-5 February 1922):** In the Gorakhpur district of the United Provinces, a peaceful procession turned violent when police fired upon the crowd. The agitators retaliated by setting the local police station on fire, resulting in the death of 22 policemen. Disturbed by this violence, Gandhi abruptly withdrew the Non-Cooperation Movement.

Q 23. Ans. B

Explanation:

- **Statement 1 is correct:** The Swaraj Party (formally the Congress-Khilafat Swarajya Party) emerged following an ideological split at the Gaya session of the Indian National Congress in December 1922. The 'Pro-changers', led by C.R. Das and Motilal Nehru, advocated contesting elections to enter and disrupt the legislative councils. When their proposal was defeated by the orthodox Gandhians ('No-changers'), they resigned from their Congress posts and formed the new party on 1 January 1923.
- **Statement 2 is incorrect:** The statement reverses the respective leadership roles. Motilal Nehru served as the leader of the Swaraj Party in the Central Legislative Assembly at Delhi. C.R. Das, holding immense regional political influence, led the party in the Bengal Provincial Legislative Council.
- **Statement 3 is correct:** The British government appointed the Muddiman Committee (Reforms Enquiry Committee) in 1924 to assess the functioning of the dyarchy system introduced by the 1919 Act. When the official majority report claimed the system was working adequately, Motilal Nehru moved a resolution in the Central Legislative Assembly rejecting this view. The Swarajists successfully passed an amendment condemning dyarchy and demanding the drafting of a new, fully responsible constitution for India.

Q 24. Ans. A

Explanation:

- **Statement 1 is incorrect:** The Poona Pact, signed in September 1932, was an agreement reached between Dr B.R. Ambedkar (representing the Depressed Classes) and upper-caste Hindu leaders (principally Madan Mohan Malaviya), not the British Government. The British Government merely accepted the agreement to formally amend the provisions of the earlier Communal Award.

- **Statement 2 is correct:** The pact successfully replaced the provision of separate electorates with joint electorates. In exchange, the number of reserved seats for the Depressed Classes in the provincial legislatures was substantially increased from 71 (initially granted by the Communal Award) to 148. It also provided for 18 percent reservation of seats in the Central Legislature.
- **Statement 3 is correct:** In August 1932, British Prime Minister Ramsay MacDonald announced the Communal Award, which extended separate electorates to the Depressed Classes. Viewing this as a deliberate imperial strategy to permanently divide Hindu society, Mahatma Gandhi commenced a fast unto death inside the Yeravada Central Jail on 20 September 1932. This extreme political pressure compelled leaders across factions to negotiate and finalise the Poona Pact.

Q 25. Ans. A

Explanation:

- **Statement 1 is correct:** The British Prime Minister, Clement Attlee, introduced the Indian Independence Bill in the House of Commons on 4 July 1947. It was swiftly passed by the British Parliament and received Royal Assent on 18 July 1947, providing the legal framework for the transfer of power.
- **Statement 2 is correct:** To execute the partition, two separate boundary commissions were established under the chairmanship of the British jurist Sir Cyril Radcliffe. One commission was tasked with demarcating the borders in Punjab, and the other was responsible for the partition of Bengal, based on contiguous Muslim and non-Muslim majority areas.
- **Statement 3 is incorrect:** Formally and legally, Section 1 of the Indian Independence Act 1947 stipulated that both the independent dominions of India and Pakistan would come into existence on the exact same date: 15 August 1947. Although Lord Mountbatten held the transfer of power ceremony in Karachi on 14 August so he could return to New Delhi for India's ceremony, the statutory date of creation for both nations remained 15 August.
- **Statement 4 is correct:** While the Muslim League chose Mohammad Ali Jinnah to be the first Governor-General of Pakistan, the Indian National Congress requested Lord Mountbatten to continue in his role. Consequently, Mountbatten served as the first Governor-General of the independent Dominion of India.

Q 26. Ans. C

Explanation:

- Aurangzeb (1658–1707) is associated with the ban on music at the imperial court as part of his more orthodox religious outlook.
- In 1668, Aurangzeb discontinued the official practice of court music and dismissed many court musicians from imperial service.
- According to tradition, when musicians staged a symbolic funeral procession for music, Aurangzeb remarked that they should “bury it so deep that no sound of it may ever rise again.”
- Despite the ban at the imperial court, music continued to flourish in regional courts and among sections of the nobility.

Q 27. Ans. D

Explanation:

- The early nationalists (Moderates) played a crucial role in introducing and popularising several modern political ideas in India. Their contribution extended beyond constitutional demands and laid the ideological foundations of the national movement.
- **Democracy**
 - ✓ They advocated representative institutions, elected legislative councils, and greater Indian participation in administration, thereby familiarising Indians with democratic principles.
- **Civil liberties**
 - ✓ They demanded freedom of speech, freedom of the press, freedom of association, and protection of individual rights, opposing arbitrary colonial restrictions.
- **Secularism**
 - ✓ Early nationalist leaders consciously promoted unity across religious communities, emphasising that Indians constituted one nation irrespective of caste, creed, or religion.
- **Nationalism**
 - ✓ Their foremost achievement was the development of an all-India national consciousness, encouraging people to think beyond provincial, caste, and religious identities.

Q 28. Ans. C

Explanation:

- **Statement 1 is correct:** Announced by Viceroy Linlithgow in August 1940, the August Offer formally recognised for the first time the inherent right of Indians to frame their own constitution.

It explicitly proposed that a representative Indian body would be established to draft the new constitution immediately after the conclusion of the Second World War.

- **Statement 2 is correct:** To register a moral protest against the British policy of dragging India into the war without consultation, Mahatma Gandhi launched the Individual Satyagraha in October 1940. He selected Acharya Vinoba Bhave as the first satyagrahi to deliver anti-war speeches and court arrest, followed by Jawaharlal Nehru as the second, and Brahma Datt as the third.
- **Statement 3 is incorrect:** The Cripps Mission (1942) failed primarily because it did not offer immediate self-government. It proposed the creation of a new Indian Union with Dominion Status, but this was strictly conditional upon the end of the war. Due to this deferred nature of the proposal during a time of crisis, Mahatma Gandhi famously rejected it as a ‘post-dated cheque on a crashing bank’.

Q 29. Ans. C

Explanation:

- The first jute mill in India was established in 1855 at Rishra, near Serampore in present-day West Bengal, on the banks of the Hooghly River.
- It was founded by George Acland, a British entrepreneur.
- The mill marked the beginning of the modern jute industry in India, which later became concentrated along the Hooghly industrial belt.
- The industry expanded rapidly due to the availability of raw jute from eastern India, abundant labour, river transport, and proximity to the port of Calcutta.

Q 30. Ans. C

Explanation:

- **Statement 1 is correct:** The Quit India Resolution was formally ratified and passed by the All India Congress Committee (AICC) during its session at the Gowalia Tank Maidan in Bombay on 8 August 1942. It was here that Mahatma Gandhi delivered his famous ‘Do or Die’ speech.
- **Statement 2 is correct:** Following the arrests of the top-tier Congress leadership on the morning of 9 August, a second-rung leadership rapidly organised an underground resistance. Key figures who successfully evaded arrest and coordinated these clandestine activities included Jayaprakash Narayan, Ram Manohar Lohia, Achyut Patwardhan, Aruna Asaf Ali, and Sucheta Kripalani.

- **Statement 3 is incorrect:** The British authorities segregated the arrested leadership to prevent communication. While Jawaharlal Nehru, Maulana Abul Kalam Azad, and other prominent members of the Congress Working Committee were incarcerated at the Ahmednagar Fort, Mahatma Gandhi, along with his wife Kasturba and secretary Mahadev Desai, was imprisoned separately at the Aga Khan Palace in Poona (Pune).

Q 31. Ans. C

Explanation:

- West Champaran (Paschim Champaran), with its headquarters at Bettiah, is the largest district of Bihar by area and shares the longest international boundary with Nepal among all districts of the state.
- The open international border with Nepal spans five blocks, which are: Bagaha-II, Ramnagar, Gaunaha, Mainatand, and Sikta, covering approximately 35 km.
- The district occupies a prominent place in Indian history as the site of the Champaran Satyagraha of 1917.
- West Champaran is also renowned for its rich ecological heritage, being home to the Valmiki Tiger Reserve, the only tiger reserve and national park in Bihar, located in the Himalayan foothill region and covering approximately 336 sq. km of core area.
- The district is economically important due to its agricultural base and the presence of several sugar mills in Majhaulia, Bagaha, and Narkatiaganj, along with numerous rice mills.
- It also possesses significant archaeological and cultural heritage, including the famous Ashokan Lion Pillar at Lauriya and the ancient burial mounds of Nandangarh, which provide valuable evidence of the Mauryan period.
- Another notable attraction is Saraiya Man Lake, a scenic horseshoe-shaped lake situated.

Q 32. Ans. A

Explanation:

- **Statement 1 is correct:** After completing his education in the United States and returning to India in 1929, Jayaprakash Narayan joined the **Indian National Congress (INC)**. During this period, he was influenced by socialist and Marxist ideas and became actively involved in India's freedom struggle. From 1929 onwards, he emerged as a prominent nationalist leader and participated in various movements against British rule.
- **Statement 2 is incorrect:** During the **Quit India Movement (1942)**, Jayaprakash Narayan

escaped from Hazaribagh Jail and continued the underground struggle against British rule. He organised an "**Azaad Dasta**" (**Freedom Brigade**) in **Nepal**, not Bhutan, to coordinate revolutionary activities against the colonial government.

- **Statement 3 is incorrect:** Jayaprakash Narayan, popularly known as **Lok Nayak**, was awarded the **Bharat Ratna posthumously in 1999**, not in 1996. The award was conferred by the Government of India in recognition of his immense contribution to the freedom struggle, socialist movement, and the movement for "**Total Revolution**" against corruption and authoritarianism.

Q 33. Ans. B

Explanation:

- The **Barahiya Taal Struggle (1936–1939)** was an important peasant movement in the **Barahiya Taal region of present-day Lakhisarai district (then part of Munger district), Bihar**. It formed a significant part of the broader **Bakasht Movement** led against the zamindari system. Under the leadership of **Karyanand Sharma** and the **Bihar Provincial Kisan Sabha**, peasants and sharecroppers (**Bataidars**) opposed the attempts of zamindars to reclaim **Bakasht lands** that tenants had been cultivating. The movement gained momentum during the **Great Depression**, when many cultivators were unable to pay rent and faced eviction. In **November 1936**, the Bihar Provincial Kisan Sabha mobilised the peasants to resist the landlords' claims and harvest the crops themselves, making the struggle a notable chapter in Bihar's agrarian and anti-zamindari movements.

Q 34. Ans. A

Explanation:

- **Pair 1 is correct:** Ajgaibinath Dham is situated at **Sultanganj in Bhagalpur district** on the bank of the Ganga River. It is one of the most important Shiva temples in Bihar. A unique feature of Sultanganj is that the **Ganga flows northward (Uttarvahini)** here, which is considered highly auspicious in Hindu tradition. The temple is closely associated with the **Shravani Mela**, during which millions of Kanwariyas collect holy Ganga water from Sultanganj and carry it approximately 105 km to the **Baidyanath Temple in Deoghar (Jharkhand)**.
- **Pair 2 is incorrect:** Budhanath Temple is **not** located in Bhawanipur, Madhubani. The temple is situated in **Jogsar locality of Adampur, Bhagalpur**.

district. It is an ancient Shiva temple and one of the major religious landmarks of Bhagalpur. The temple attracts large numbers of devotees, especially during Mahashivaratri and the Shravani festivities. **Bhawanipur, Madhubani**, is actually associated with the **Ugna Mahadev Temple**, not Budhanath Temple.

- **Pair 3 is incorrect:** Harihar Nath Temple is not located in Adampur, Bhagalpur. The temple is situated at **Sonepur (Harihar Kshetra) in Saran district**, near the confluence of the **Ganga and Gandak rivers**. The deity represents the combined form of **Hari (Vishnu) and Har (Shiva)**, making it an important center of both Shaivite and Vaishnavite worship. The temple is internationally known because of the famous **Sonepur Mela**, one of Asia's largest traditional fairs.
- **Pair 4 is incorrect:** Ugna Mahadev Temple is not situated in Sonepur. It is located in **Bhawanipur village of Madhubani district** and is deeply associated with the celebrated Maithili poet **Vidyapati**. According to local tradition, Lord Shiva appeared before Vidyapati in the form of a servant named **Ugna** and served him for many years. The temple commemorates this legendary association and is an important center of Maithili cultural heritage. Since Sonepur is the location of **Harihar Nath Temple**, not Ugna Mahadev Temple, the pair is incorrect.

Q 35. Ans. C

Explanation:

- **BHAVYA** (Bihar Health Application Visionary Yojana for All) is a flagship healthcare digitization initiative of the Health Department, Government of Bihar, aimed at transforming public health service delivery through a comprehensive digital ecosystem.
- Under the programme, a unified Hospital Information Management System (HIMS) is being implemented across government health institutions to streamline healthcare operations. The platform integrates nearly 13,000 public health facilities and connects more than 30,000 healthcare professionals across the state. Through its citizen-centric portal, patients can register online, schedule doctor appointments, access laboratory reports, and locate nearby blood banks. BHAVYA also enables end-to-end digital management of healthcare services, covering OPD and IPD registration, pharmacy medicine dispensing, emergency care, diagnostic services, and billing, thereby enhancing efficiency, transparency, and accessibility in Bihar's healthcare system.

Q 36. Ans. D

Explanation:

- Banabhatta was a distinguished **7th-century Sanskrit scholar, poet, and prose writer**, best known as the **court poet of Emperor Harshavardhana**. His most celebrated work, **Kadambari**, is regarded as one of the earliest novels in world literature, presenting a highly imaginative narrative centered on themes of love, romance, and rebirth across multiple lifetimes. Another notable work, **Harshacharita**, is considered the **first historical biography in Sanskrit**, offering a detailed and literary account of the life and reign of Emperor Harsha in eight chapters. His other important compositions include **Chandikashataka**, a devotional hymn consisting of one hundred verses dedicated to Goddess Chandi, and **Parvatiparinaya**, a dramatic work depicting the marriage of Goddess Parvati and Lord Shiva.
- Vidyapati (c. 1352–1448), revered as the “**Maithil Kavi Kokil**” and the “**Adikavi**” of **Maithili literature**, was one of the greatest poets of medieval eastern India. Writing in both **Maithili and Sanskrit**, he elevated Maithili into a prestigious literary language. His most famous work, **Vidyapati Padavali**, is a celebrated collection of lyrical poems depicting the divine love of Radha and Krishna, while also portraying seasonal beauty and local cultural traditions. Beyond devotional poetry, Vidyapati authored several important socio-political and philosophical works, including **Purusha Pariksha**, **Bhuparikrama** (a geographical and travel-oriented work), **Danavakyavali** (dealing with charity and ethics), and **Saubhagyodaya**.
- Phanishwar Nath 'Renu' was a pioneering figure of the **Aanchalik (regional) literary movement** in modern Hindi literature. His masterpiece, **Maila Anchal** (1954), is widely regarded as one of the greatest Hindi novels and is often considered the most significant Hindi novel after Premchand's Godaan. Other notable works include **Parti Parikatha**, which explores rural life and social transformation; **Juloos**, depicting social and political dynamics in village society; **Kitne Chaurahe**, reflecting the complexities of changing social values; and **Paltu Babu Road**, a satirical novel centered on life in a small town.
- Jyotirishwar Thakur was a renowned **Maithil scholar, priest, and poet** of the 14th century. He is best known for authoring **Varna Ratnakara** (1324 CE), the **oldest extant prose work in the**

Maithili language. Literally meaning “Ocean of Description,” the work is an encyclopedic literary composition that provides detailed descriptions of contemporary society, culture, occupations, customs, and various aspects of life in medieval Mithila.

Q 37. Ans. D

Explanation:

- **Statement:** Bihar’s own tax revenue has witnessed substantial growth over the years, including in the post-GST period. The increase reflects a broader tax base, higher economic activity, improved compliance, and more efficient tax administration and collection mechanisms within the state.
- **Conclusion I:** Economic activity and tax administration have improved. An increase in own tax revenue is a reasonable indicator of improved economic activity and more efficient tax administration. Therefore, this conclusion logically follows from the statement.
- **Conclusion II:** Bihar has become financially independent of Central transfers is incorrect. Bihar continues to receive a substantial share of its revenue through **tax devolution, Finance Commission grants, and Centrally Sponsored Schemes**. Hence, financial independence from Central transfers cannot be inferred from the statement.

Q 38. Ans. B

Explanation:

- **Statement 1 is correct:** In a significant reform of the electoral process, the **Bihar State Election Commission (SEC)** decided to introduce an **Android-based e-voting system**. The entire e-voting process is conducted through two mobile applications, one of which is “**e-Voting SECBHR**”. This application was developed specifically for the implementation of remote electronic voting. The system incorporates advanced technologies such as **blockchain architecture, liveness detection, face matching, live face scanning, and face comparison** to ensure secure and tamper-proof voting. Similar to the **VVPAT mechanism used with EVMs**, an audit trail has also been integrated into the e-voting process.
- **Statement 2 is incorrect:** The Bihar State Election Commission described the initiative as a **paradigm shift in the voting process** and stated that **Bihar would become the first state in India** to adopt such an Android-based e-voting system.

- **Statement 3 is correct:** According to the State Election Commissioner, based on the information available on e-voting systems globally, **Estonia** is a country that has successfully implemented an **e-voting process**. Estonia is widely recognised for pioneering internet-based voting and is often cited as a leading example of digital electoral innovation.

Q 39. Ans. C

Explanation:

- The Bihar Police has been conferred the prestigious President’s Police Colour Award recently, it is regarded as the highest honour bestowed upon a police organisation in India for outstanding service, exemplary courage, and unwavering dedication to duty. Also known as the “Nishan”, the award recognizes the collective achievements and distinguished record of a police force. In the same year, the police forces of Odisha and Sikkim were also accorded this distinction. As part of the honour, Bihar Police personnel—from constables to the Director General of Police (DGP)—will be entitled to wear a special insignia (flag badge) on the left sleeve of their uniform, symbolizing the force’s exceptional contribution to policing and public service.

Q 40. Ans. D

Explanation:

- **Statement 1 is incorrect:** The Fourth Agriculture Road Map (2023–2028) of Bihar was launched by the President of India, Smt Droupadi Murmu, on October 18, 2023, at Patna, not in 2025. The roadmap was introduced as a comprehensive strategy to accelerate agricultural growth, improve farmers’ income, and promote sustainable agricultural development in the state.
- **Statement 2 is incorrect:** The statement is incorrect on two counts. First, the Fourth Agriculture Road Map covers a five-year period (2023–2028), not four years. Second, the description of a “landmark strategic blueprint designed to revolutionize the agrarian sector” specifically relates to Bihar’s First Agriculture Road Map (2008–2012), which was launched on 17 February 2008. The first roadmap marked a major shift from traditional farming practices towards a more modern, sustainable, and profitable agricultural framework.
- **Statement 3 is incorrect:** The Fourth Agriculture Road Map has a much larger financial outlay of approximately ₹1.62 lakh crore (₹1,62,268

crore). Furthermore, it is a comprehensive, multi-sectoral initiative that is implemented through the coordinated efforts of 12 different state departments (including Agriculture, Water Resources, Minor Water Resources, Animal and Fisheries Resources, Cooperative, etc.), not just the Agriculture Department along with 7 others.

Q 41. Ans. B

Explanation:

- **Harishchandra Range** → Stretches across Pune and Osmanabad districts of Maharashtra
 - ✓ The Harishchandra Range is a prominent range of the Western Ghats in Maharashtra. It extends through parts of the state, including the regions associated with Pune and Osmanabad.
- **Kudremukh** → Rich in haematite and magnetite iron ore
 - ✓ Located in Karnataka, Kudremukh is renowned for its extensive deposits of high-grade haematite and magnetite iron ore. The region was historically important for iron ore mining.
- **Mahabaleshwar** → An important peak of the Western Ghats at 1438 m
 - ✓ Mahabaleshwar, situated in Maharashtra, is a major hill station and one of the important elevated points of the Western Ghats, with an elevation of about 1,438 metres.
- **Nilgiri Hills** → Famous hill resort of southern India
 - ✓ The Nilgiri Hills, located at the junction of Tamil Nadu, Kerala, and Karnataka, are known for celebrated hill stations such as Ooty (Udhagamandalam), making them one of the most famous hill resort regions of southern India.

Q 42. Ans. D

Explanation:

- **Amw** → Tropical Monsoon Climate
 - ✓ Amw denotes the Tropical Monsoon Climate in Köppen's classification. It is characterised by heavy seasonal rainfall and a short dry season. It does not refer to "Tropical Moist (dry summers)."
- **As** → Tropical Savanna Climate with dry summer season
 - ✓ As represents the Tropical Savanna Climate with a dry summer, not a semi-arid steppe climate. Semi-arid steppe climates are classified under BS.

- **BShw** → Semi-arid Steppe Climate (hot and dry, with dry winter)

✓ BShw denotes a Hot Semi-arid (Steppe) Climate with dry winters. It does not indicate a cold humid winter climate.

- **Dfc** → Cold climate with no dry season and cool summer

✓ Dfc refers to a Subarctic Climate with cool summers and no dry season, not a tropical monsoon climate.

Q 43. Ans. C

Explanation:

- The Bedouins are traditionally nomadic Arab pastoralists who inhabit the vast desert regions of West Asia (including the Arabian Peninsula, the Levant, and Mesopotamia) and North Africa (including the Sahara and Egypt).
- Historically, their socio-economic survival depended entirely on pastoralism, specifically the herding of camels, sheep, and goats. They lived in tents and continuously migrated across arid landscapes in search of scarce water sources and seasonal grazing pastures.
- Their entire culture, from their loose-fitting clothing designed to protect against extreme heat and sand, to their diet and tribal governance, is a direct evolutionary adaptation to the harsh, unforgiving climate of the desert.
- The Arctic region is home to groups like the Inuit; the Amazon rainforest is inhabited by indigenous tribes such as the Yanomami; and the grasslands of South America (the Pampas) are historically associated with the Gauchos.

Q 44. Ans. A

Explanation:

- The Oder-Neisse Line is the internationally recognised border between Germany and Poland. It was established in the aftermath of the Second World War during the Potsdam Conference in 1945.
- The boundary is primarily defined by the courses of the Oder River and its tributary, the Lusatian Neisse River, running from the Baltic Sea in the north to the Czech Republic border in the south.
- The implementation of this line resulted in a major westward shift of Poland's borders. It transferred significant pre-war eastern German territories, such as Silesia, Pomerania, and parts of East Prussia, to Polish and Soviet administration.

● **Other prominent international boundary lines:**

- ✓ **Radcliffe Line:** Demarcates the boundary between India and Pakistan (which included present-day Bangladesh), drawn by Sir Cyril Radcliffe during the partition in 1947.
- ✓ **MacMahon Line:** Forms the boundary between India and China (specifically the Tibetan region) in the eastern Himalayan sector, negotiated at the Simla Convention in 1914.
- ✓ **Durand Line:** The international border separating Pakistan and Afghanistan, established by Sir Mortimer Durand in 1893.
- ✓ **Curzon Line:** Proposed by British Foreign Secretary Lord Curzon in 1920, it roughly forms the modern eastern border of Poland, separating it from Belarus and Ukraine.
- ✓ **Maginot Line:** A heavily fortified defensive boundary built by France in the 1930s along its border with Germany to deter invasion.
- ✓ **Siegfried Line:** The opposing German defensive line built parallel to the French Maginot Line.
- ✓ **Mannerheim Line:** A defensive fortification line built by Finland across the Karelian Isthmus to protect against the Soviet Union.
- ✓ **38th Parallel:** The latitudinal line that serves as the border separating North Korea and South Korea, established post-World War II.
- ✓ **49th Parallel:** Forms the vast majority of the international border between the United States of America and Canada.
- ✓ **17th Parallel:** The historical boundary that temporarily partitioned North Vietnam and South Vietnam following the 1954 Geneva Accords.
- ✓ **24th Parallel:** A boundary line in the Rann of Kutch claimed by Pakistan as the border with India, which India actively rejects.

Q 45. Ans. B

Explanation:

- The Dardanelles, historically known as the Hellespont, is a narrow, natural strait situated in northwestern Turkey. Along with the Bosphorus, it forms parts of the continental boundary separating Europe and Asia.
- It serves as a vital maritime passage that connects the Sea of Marmara to the Aegean Sea, providing a route to the broader Mediterranean Sea.
- This strait is of immense geopolitical and commercial significance. It is a mandatory transit

route for naval and commercial vessels travelling from the Black Sea out to the global oceans.

● **Other prominent international straits:**

- ✓ **Bosphorus Strait:** Connects the Black Sea and the Sea of Marmara in Turkey.
- ✓ **Strait of Gibraltar:** Connects the Mediterranean Sea and the Atlantic Ocean, separating Spain and Morocco.
- ✓ **Strait of Hormuz:** Connects the Persian Gulf and the Gulf of Oman, serving as a critical global choke point for oil transit.
- ✓ **Bab-el-Mandeb:** Connects the Red Sea and the Gulf of Aden, separating the Arabian Peninsula (Yemen) from the Horn of Africa (Djibouti and Eritrea).
- ✓ **Strait of Malacca:** Connects the Andaman Sea (Indian Ocean) and the South China Sea (Pacific Ocean), separating the Malay Peninsula from the Indonesian island of Sumatra.
- ✓ **Bering Strait:** Connects the Arctic Ocean and the Bering Sea, geographically separating Asia (Russia) from North America (Alaska, USA).
- ✓ **Palk Strait:** Connects the Bay of Bengal and the Palk Bay, separating the southeastern coast of India from Sri Lanka.
- ✓ **Strait of Magellan:** Connects the Atlantic and Pacific Oceans, acting as a natural navigable route separating mainland South America from Tierra del Fuego.
- ✓ **Strait of Tartary (Tatar Strait):** Connects the Sea of Okhotsk and the Sea of Japan, separating the Russian island of Sakhalin from mainland Asia.

Q 46. Ans. A

Explanation:

- The Zero Mile Stone is a historic monument located in the geographical center of mainland India in the city of Nagpur, Maharashtra.
- It was erected in 1927 by the British during the colonial period.
- The British established this monument to serve as the principal reference point (the “zero mile”) for measuring geographical distances to all other major cities and provinces across the Indian subcontinent. It was a crucial marker for the Great Trigonometrical Survey of India.
- The monument features a sandstone pillar indicating the exact central coordinate. It is accompanied by four stucco horses added later, which are positioned to face the four cardinal directions (North, South, East, and West).

Q 47. Ans. C

Explanation:

- The name “Rub al Khali” translates directly from Arabic to “Empty Quarter” in English. It earns this name because its extremely harsh environment makes it almost entirely uninhabitable for humans.
- **It is situated in the southern portion of the Arabian Peninsula. The desert spans across four countries:** primarily Saudi Arabia, as well as Oman, the United Arab Emirates (UAE), and Yemen.
- The Rub al Khali is the largest continuous sand desert (or erg) in the world. It is characterised by massive, towering sand dunes, hyper-arid climate conditions, and virtually no surface water.
- **Other prominent global deserts:**
 - ✓ **Sahara Desert:** The largest hot desert in the world, stretching across most of North Africa.
 - ✓ **Gobi Desert:** A massive cold winter desert located in East Asia, covering parts of Northern China and Southern Mongolia. It is notable for being a rain shadow desert caused by the Tibetan Plateau.
 - ✓ **Karakum Desert:** Located primarily in Turkmenistan (Central Asia). Its name translates to “Black Sand,” and it is famous for the Darvaza Gas Crater, often called the “Door to Hell.”
 - ✓ **Atacama Desert:** Located in South America (primarily Chile), it is recognised as the driest non-polar desert in the world.
 - ✓ **Taklamakan Desert:** A cold desert in the Xinjiang region of China. Its name is popularly interpreted to mean “once you go in, you will never come out.”
 - ✓ **Kalahari Desert:** A large semi-arid sandy savanna in Southern Africa, extending across Botswana, Namibia, and South Africa.
 - ✓ **Thar Desert:** Also known as the Great Indian Desert, it forms a natural geopolitical boundary between northwestern India and eastern Pakistan.
 - ✓ **Patagonian Desert:** The largest desert in Argentina and the eighth-largest in the world, primarily a cold winter desert.

Q 48. Ans. D

Explanation:

- The ‘Taungya’ system of cultivation originated in Myanmar (historically known as Burma) in the mid-19th century. In the Burmese language, taung translates to “hill” and ya translates to “cultivation”.

- It is a traditional form of agroforestry. Under this system, landless farmers are temporarily allocated plots of land to cultivate agricultural crops. Simultaneously, they are required to plant and tend to forestry tree seedlings (often teak) on the same plot.
- Once the forest trees grow large enough that their canopy closes and blocks the sunlight, making agricultural farming impossible, the farmers move to a new, cleared site to repeat the process. This system historically allowed the state to establish timber plantations at a very low cost while providing farmers with temporary farming land.
- **Other prominent traditional/shifting cultivation systems:**
 - ✓ **Ladang:** Indonesia and Malaysia
 - ✓ **Roca:** Brazil
 - ✓ **Milpa:** Mexico and Central America
 - ✓ **Chena:** Sri Lanka
 - ✓ **Masole:** Congo Basin (Central Africa)
 - ✓ **Ray:** Vietnam
 - ✓ **Tavy:** Madagascar
 - ✓ **Caingin:** Philippines
 - ✓ **Jhum (or Jhuming):** North-Eastern states of India
 - ✓ **Podu:** Andhra Pradesh and Odisha, India
 - ✓ **Bewar or Dahiya:** Madhya Pradesh, India

Q 49. Ans. B

Explanation:

- Lake Superior is the largest freshwater lake in the world by surface area, covering approximately 82,100 square kilometers. It is the largest of the five Great Lakes of North America and is shared by the United States and Canada.
- While Lake Baikal in Siberia, Russia, is not the largest by surface area, it holds the record for being the world’s largest freshwater lake by volume and the deepest lake in the world. It contains roughly 20% of the world’s unfrozen surface fresh water.
- The Caspian Sea is the world’s largest inland body of water (lake) by surface area and volume overall. However, it is a saltwater lake.
- Located in East Africa, Lake Victoria is the world’s second-largest freshwater lake by surface area and the largest lake on the African continent.
- **Other prominent global lake records:**
 - ✓ **Lake Tanganyika:** The world’s longest freshwater lake, and the second-largest by volume and second-deepest (after Lake Baikal). It is located in the Great Rift Valley of Africa.
 - ✓ **Lake Titicaca:** The highest commercially navigable

lake in the world, situated high in the Andes mountains on the border of Bolivia and Peru.

- ✓ **Dead Sea:** A salt lake bordered by Jordan to the east and Israel and the West Bank to the west, possessing the lowest land elevation on Earth's surface and ranking among the world's saltiest bodies of water.
- ✓ **Lake Vostok:** The largest known subglacial lake in the world, buried beneath nearly 4 kilometers of ice in Antarctica.
- ✓ **Lake Huron and Lake Michigan:** Part of the North American Great Lakes. Technically, they are functionally a single lake (Lake Michigan-Huron) connected by the Straits of Mackinac.

Q 50. Ans. C

Explanation:

- The Godavari basin is the largest river basin in Peninsular India, covering an area of nearly 3.13 lakh square kilometers, which is almost 10% of the total geographical area of the country.
- The Godavari river originates from the Brahmagiri mountains at Trimbakeshwar in the Nashik district of Maharashtra and flows eastwards to drain into the Bay of Bengal.
- The extensive basin encompasses major parts of the states of Maharashtra, Telangana, Andhra Pradesh, Chhattisgarh, and Odisha, with smaller catchment areas falling in Madhya Pradesh, Karnataka, and the Union Territory of Puducherry (Yanam).
- Due to its immense length (approx. 1,465 km) and vast catchment area, the Godavari is frequently referred to as the 'Dakshin Ganga' (Ganges of the South) or 'Vridha Ganga' (Old Ganges).
- **Other prominent Peninsular rivers:**
 - ✓ **Krishna River:** The second-largest east-flowing Peninsular river basin. It originates near Mahabaleshwar in Maharashtra and is known for major tributaries like the Tungabhadra, Bhima, and Musi.
 - ✓ **Mahanadi River:** Originates in the highlands of Chhattisgarh and flows eastwards through Odisha. It is famous for the Hirakud Dam, which is the longest earthen dam in the world.
 - ✓ **Kaveri (Cauvery) River:** Originates from Talakaveri in the Brahmagiri hills of Karnataka. Uniquely among Peninsular rivers, it has a relatively perennial flow because its upper catchment area receives rainfall from the South-West monsoon (summer), while the lower part receives rainfall from the North-East monsoon (winter).

- ✓ **Narmada River:** The largest west-flowing Peninsular river. It originates from the Amarkantak plateau in Madhya Pradesh and uniquely flows through a rift valley situated between the Vindhya and Satpura mountain ranges before draining into the Arabian Sea.
- ✓ **Tapi (Tapti) River:** The second-largest west-flowing Peninsular river. Like the Narmada, it also flows through a rift valley, running parallel and to the south of the Satpura range.

Q 51. Ans. A

Explanation:

- Karnaprayag is the specific geographical location in the Garhwal Himalayas of Uttarakhand where the Alaknanda River is joined by the Pindar River (also known as the Pindari River).
- The Alaknanda originates from the Satopanth and Bhagirath Kharak glaciers, while the Pindar River originates from the Pindari Glacier located in the Kumaon region.
- **Other prominent confluences (The Panch Prayag):**
 - ✓ **Vishnuprayag:** The confluence of the Alaknanda River and the Dhauliganga River.
 - ✓ **Nandaprayag:** The confluence of the Alaknanda River and the Nandakini River.
 - ✓ **Karnaprayag:** The confluence of the Alaknanda River and the Pindar River.
 - ✓ **Rudraprayag:** The confluence of the Alaknanda River and the Mandakini River.
 - ✓ **Devprayag:** The final and most significant confluence, where the Alaknanda River meets the Bhagirathi River. It is exclusively from this point onwards that the combined river officially takes the name Ganga (Ganges).

Q 52. Ans. A

Explanation:

- **Mana Pass → Connects Uttarakhand to Tibet**
 - ✓ Mana Pass is one of the highest motorable passes in India, located in Uttarakhand's Chamoli district. It historically served as an important trade route connecting India with Tibet.
- **Nathu La → Located in Sikkim border posts between India and China**
 - ✓ Nathu La Pass, situated in East Sikkim, is a strategically important pass on the India-China border and forms part of the historic Silk Route.
- **Pensi La → Connects the Suru Valley to the Zaskar Valley**

- ✓ Pensi La is a high mountain pass located at an elevation of 4,400 metres in the Ladakh Union Territory, situated on the NH301 Kargil–Padum Highway. It connects the Suru Valley in Kargil district to the Zaskar Valley and is widely known as the Gateway to Zaskar. The pass is also notable for its proximity to the Drang Drung Glacier, one of the largest glaciers in Ladakh outside the Karakoram range.
- **Qara Tagh Pass → Located in the Karakoram on the Leh–Yarkand caravan route**
 - ✓ The Qara Tagh Pass lies in the Karakoram Range and was historically used on the famous Leh–Yarkand trade route linking Ladakh with Central Asia.

Q 53. Ans. D

Explanation:

- **Kothagudem Thermal Power Station → Telangana**
 - ✓ Located in the Bhadrachalam Kothagudem district of Telangana, it is one of the major thermal power stations operated by the Telangana Power Generation Corporation (TGGENCO).
- **Korba Thermal Power Station → Chhattisgarh**
 - ✓ The Korba region in Chhattisgarh is a major power-producing hub and houses several thermal power plants, including the NTPC Korba Super Thermal Power Station.
- **Vindhyachal Super Thermal Power Station → Madhya Pradesh**
 - ✓ Situated in the Singrauli district of Madhya Pradesh, Vindhyachal is one of India's largest thermal power stations and is operated by NTPC.
- **Koradi Thermal Power Station → Maharashtra**
 - ✓ The Koradi Thermal Power Station is located near Nagpur in Maharashtra and is operated by the Maharashtra State Power Generation Company (MAHAGENCO).

Q 54. Ans. C

Explanation:

- According to the Global Hunger Index (GHI) 2025 (and consistently reported in the 2023 and 2024 editions), India's child wasting rate stands at 18.7%. This represents the highest child wasting rate globally.
- Child wasting is defined as the proportion of children under the age of five who have a low weight for their height. It is a critical health metric indicating acute undernutrition, often resulting from insufficient caloric intake, poor sanitation, or severe illnesses.

- **Global Hunger Index calculates a country's score using four specific indicators:** child wasting (acute undernutrition), child stunting (chronic undernutrition, measured by low height for age), undernourishment (insufficient caloric intake across the population), and child mortality (children who die before their fifth birthday).
- The Ministry of Women and Child Development has pointed out that real-time data from its POSHAN Tracker application consistently records a much lower child wasting rate, typically below 7.2%.
- To combat malnutrition, food insecurity, and poor maternal health, the Indian government heavily relies on targeted welfare initiatives. Major programs include POSHAN Abhiyaan (National Nutrition Mission), the Integrated Child Development Services (ICDS), the Mid-Day Meal Scheme, and the distribution of subsidised food grains via the National Food Security Act (NFSA)

Key Indicator / Topic	2025 Value & Observations
Global Rank	102 out of 123 countries
Overall GHI Score	25.8
Hunger Severity Category	Serious
Child Stunting (< 5 years)	32.9% (Remains extremely alarming, affecting ~1/3 of children)
Child Wasting (< 5 years)	18.7% (Improved slightly but remains extremely alarming)
Undernourished Population	12.0% (Approx. 172 million people; 13.5 million higher than in 2016)
Child Mortality (< 5 years)	2.8%
South Asia Comparison	India (25.8) is slightly worse than the South Asian average (24.9)
Global Comparison	India (25.8) is worse than the World average (18.3)

Q 55. Ans. B

Explanation:

- A Giffen good is a rare type of non-luxury item whose demand increases as its price rises, defying standard economic principles.
- Giffen goods defy the traditional law of demand by increasing in demand as prices rise.

- Named after Sir Robert Giffen, these goods are typically essential, low-income items with few substitutes.
- The upward-sloping demand curve of Giffen goods is driven by a significant income effect that outweighs the substitution effect

Q 56. Ans. C

Explanation:

- **Statement 1 is correct:** The Social Progress Index (SPI) is an annual report published by the Social Progress Imperative, a non-profit organization based in the United States.
- **Statement 2 is correct:** The index provides a comprehensive framework for measuring multiple facets of social progress across three broad dimensions: Basic Human Needs, Foundations of Wellbeing, and Opportunity.
- **Statement 3 is incorrect:** A defining and unique feature of the Social Progress Index is that it strictly measures social and environmental outcomes directly, completely excluding economic indicators such as Gross Domestic Product (GDP) or per capita income. It is designed to be independent of economic measures to show how well societies translate their economic wealth into actual social progress.
- **Statement 4 is correct:** The "Opportunity" dimension evaluates whether citizens have personal rights, freedom, and the ability to reach their full potential. Its core components include Personal Rights, Personal Freedom and Choice, Inclusiveness, and Access to Advanced Education.

Q 57. Ans. A

Explanation:

- The government-imposed upper limit on the price of a good or service is called a price ceiling.
 - ✓ It is set below the equilibrium price to make essential goods affordable.
 - ✓ **Example:** Control on prices of rent, medicines, or essential food items during shortages.

Q 58. Ans. C

Explanation:

- **Statement A is correct:** India's merchandise trade deficit during FY 2025-26 widened to US\$ 333.20 billion, representing a 17.53% increase compared to the US\$ 283.50 billion deficit recorded in FY 2024-25.
- **Statement B is correct:** Petroleum crude and products historically remain the single largest import category and the dominant driver of India's

trade deficit. This remained true even though the specific value of petroleum imports saw a 6.4% year-on-year decline in FY 2025-26.

- **Statement C is incorrect:** Gold imports did not show a significant decline; rather, they expanded significantly in value during FY 2025-26, rising by 24.1% to reach US\$ 71.98 billion. While the actual physical volume (quantity) of gold imported fell slightly from 757 tons to 721 tons, the increase in global prices pushed the overall import bill higher.
- **Statement D is correct:** Total merchandise imports for India during FY 2025-26 stood approximately at US\$ 775 billion.

Q 59. Ans. D

Explanation:

- **United States:** The USA consistently remains India's largest export destination. The market has strong demand across multiple industries, with major Indian exports including pharmaceuticals, textiles, engineering goods, IT services, and jewelry.
- **United Arab Emirates (UAE):** The UAE is a key trading partner and a leading export destination for India. It acts as a vital re-export hub to Middle Eastern and African markets, importing goods such as gold, textiles, machinery, and food products.
- **China:** Despite broad market competition, China consistently ranks among India's top five export destinations. Its large manufacturing sector creates continuous demand for significant volumes of Indian raw materials, intermediate goods, chemicals, agricultural products, and iron ore.
- **Germany:** While Germany is an important trading partner within the European Union, it is **not** among the top five overall merchandise export destinations for India. Top positions are consistently held by countries such as the United States, the UAE, the Netherlands, Singapore, and China.

Q 60. Ans. C

Explanation:

- The Reserve Bank of India (RBI) publishes the Financial Inclusion Index (FI-Index) annually to capture the depth and extent of financial inclusion across the country. According to official RBI data released in July 2025, the FI-Index value for March 2024 stood at 64.2 and increased to 67.0 in March 2025, marking a steady growth of 4.3% over the fiscal year
- **The FI-Index is constructed using 97 indicators and comprises three broad, weighted parameters that evaluate different aspects of financial engagement:**

- ✓ **Access (35% weightage):** Measures the physical and digital availability of financial services (such as bank branches, ATMs, postal services, and digital access points).
- ✓ **Usage (45% weightage):** Evaluates how actively these services are used, looking at regular account transactions, digital payments (like UPI), credit utilization, and the purchase of insurance or pension products.
- ✓ **Quality (20% weightage):** A unique and crucial feature of the index that evaluates the depth of financial inclusion by measuring financial literacy, consumer protection, grievance redressal, and service inequalities.
- While all three sub-indices showed growth, the RBI noted that the improvement in the March 2025 index was predominantly driven by significant gains in the Usage and Quality dimensions. This indicates a shift from merely opening accounts to ensuring deeper engagement with financial products and the positive impact of national financial literacy initiatives.
- The index is measured on a scale of 0 to 100, where 0 indicates complete financial exclusion and 100 represents complete financial inclusion.
- The FI-Index was constructed without a base year. This was done intentionally to reflect the cumulative, absolute progress made in financial inclusion over time.
- It incorporates data not just from banking, but across all formal financial sectors, including investments (SEBI), insurance (IRDAI), pensions (PFRDA), and postal services.
- The index was first conceptualised and officially launched in August 2021. For historical comparison, the value stood at 53.9 in March 2021, meaning the index has witnessed an absolute growth of over 24% since its inception.

Q 61. Ans. B

Explanation:

- The government has set a fiscal deficit target of **4.3% of GDP** for 2026-27. This is a deliberate reduction and an improvement over the revised estimate of **4.4% of GDP** for the previous year (2025-26), staying in line with the government's fiscal consolidation path.
- The total expenditure budgeted for 2026-27 is **₹53.47 lakh crore**. This is actually **7.7% higher** than the revised estimate of ₹49.6 lakh crore for 2025-26.
- Capital expenditure has not been reduced; it

continues its upward trajectory. The capital expenditure for 2026-27 is budgeted at **₹12.22 lakh crore**, which is an **11.5% increase** over the revised estimate of ₹10.96 lakh crore (approximately ₹11.0 lakh crore) for 2025-26, reflecting a continued push for public infrastructure investment.

Q 62. Ans. C

Explanation:

- **Statement 1 is correct:** The deposits accepted by a bank are recorded as liabilities in its balance sheet. Although deposits represent funds received by the bank, they are amounts that the bank is legally obligated to repay to depositors on demand or at maturity. Therefore, from the bank's perspective, deposits constitute liabilities.
- **Statement 2 is correct:** A bank's net worth (or shareholders' equity) is obtained by deducting total liabilities from total assets:
 - ✓ $\text{Net Worth} = \text{Total Assets} - \text{Total Liabilities}$
- Net worth represents the residual claim of the owners of the bank after meeting all its obligations. It includes components such as paid-up capital, reserves, and retained earnings.

Q 63. Ans. A

Explanation:

- When demand does not change significantly even after a large change in price, it is called inelastic demand.
 - ✓ In this case, consumers are less sensitive to price changes.
 - ✓ Essential goods like salt, medicines, or basic necessities often show inelastic demand.

Q 64. Ans. A

Explanation:

- **Statement 1 is correct:** The Pradhan Mantri Dhan-Dhaanya Krishi Yojana (PMDDKY) is strategically modelled after NITI Aayog's highly successful Aspirational Districts Programme. It targets exactly 100 underperforming agricultural districts, identified by metrics like low crop productivity, moderate cropping intensity, and below-average credit access, while ensuring that at least one district is selected from every state and union territory to maintain balanced regional development.
- **Statement 2 is correct:** A defining feature of PMDDKY is its "saturation-based convergence" approach. Rather than operating in isolation, it acts as an umbrella initiative integrating 36

active central agricultural schemes spanning 11 different ministries and departments. Through this coordinated effort, the scheme aims to directly support approximately 1.7 crore farmers, the vast majority of whom are small and marginal farmers owning less than 2 hectares of land.

- **Statement 3 is incorrect:** The scheme is approved for a **six-year period** (running from FY 2025-26 to 2030-31), not a five-year period. Furthermore, the financial outlay of ₹24,000 crore annually is divided specifically: roughly 40% is earmarked for subsidies, **30% for agricultural infrastructure** (such as post-harvest storage and micro-irrigation), 20% for agricultural loans/credit, and the remaining 10% for training and market support.

Q 65. Ans. D

Explanation:

- **Statement A is correct:** The ASI is conducted annually by the National Statistical Office (NSO) under MoSPI. It is mandated under the statutory provisions of the Collection of Statistics Act, 2008 (and its 2017 amendment, which extended the jurisdiction of the Act to include Jammu and Kashmir).
- **Statement B is correct:** The survey covers factories registered under the Factories Act, 1948 (employing 10+ workers with power, or 20+ without power). However, specific departmental units and establishments like defence establishments, railway workshops, government mints, oil storage depots, and water supply units are explicitly excluded from its purview.
- **Statement C is correct:** Maharashtra accounted for the highest share of Gross Value Added (GVA) at 15.95%, making it the top-contributing state, closely followed by Gujarat (14.20%) and Tamil Nadu (10.26%).
- **Statement D is incorrect:** According to the Annual Survey of Industries (ASI) 2023-24 results released by the Ministry of Statistics and Programme Implementation (MoSPI), textiles and petroleum products are **not** among the top five. The five largest industry contributors to Gross Value Added (GVA) are:
 1. **Basic Metals** (11.56%)
 2. **Motor Vehicles, Trailers & Semi-Trailers** (9.11%)
 3. **Chemicals & Chemical Products** (8.81%)
 4. **Food Products** (7.40%)
 5. **Pharmaceuticals, Medicinal Chemical and Botanical Products** (7.24%)

Q 66. Ans. A

Explanation:

- The primary goal of the NISAR (NASA-ISRO Synthetic Aperture Radar) mission is to make global measurements of the causes and consequences of land surface changes using advanced radar imaging. It tracks subtle changes in the Earth's crust, ice sheets, and ecosystems to understand climate change, natural hazards, and the carbon cycle.
- NISAR is a joint Earth-observing satellite mission between NASA (National Aeronautics and Space Administration, USA) and ISRO (Indian Space Research Organisation).
- It is the first satellite mission to use two different radar frequencies (L-band and S-band) to measure changes in our planet's surface less than a centimeter across. This allows it to penetrate clouds and dense forest canopies, providing data day and night, in any weather.

Q 67. Ans. D

Explanation:

- **Statement 1 is correct:** TRIGUN is the Indian Navy's indigenous air defence command and control network. It functions as the core digital nervous system for the Navy's combat and surveillance assets.
- **Statement 2 is correct:** TRIGUN is designed to seamlessly interface and operate in unison with the equivalent networks of the other armed forces. Specifically, it connects with the Indian Army's **Akashteer** (Air Defence Control and Reporting System) and the Indian Air Force's **IACCS** (Integrated Air Command and Control System).
- **Statement 3 is correct:** By aggregating and linking data from naval ship-based sensors, civilian radars, and military systems across the land and air domains, TRIGUN contributes to a unified, real-time "common operational air picture." This tri-service integration significantly improves situational awareness across the battlefield and reduces the risk of friendly fire.

Q 68. Ans. B

Explanation:

- **Statement A is incorrect:** Because of its massive weight, it was launched aboard India's heaviest and most powerful rocket, the **LVM3** (Launch Vehicle Mark-3, specifically the LVM3-M5 mission), not the much smaller PSLV.

- **Statement B is correct:** Weighing approximately 4,410 kg, CMS-03 (also known as GSAT-7R) holds the record for the heaviest communication satellite ever launched from India. While ISRO has built heavier communication satellites in the past (such as GSAT-11 at 5,854 kg and GSAT-20 at 4,700 kg), those had to be launched using foreign commercial rockets (Arianespace and SpaceX, respectively) because they exceeded ISRO's domestic payload capacities at the time.
- **Statement C is incorrect:** The rocket did not place the satellite directly into its final geostationary orbit. As is standard for heavy payloads, the LVM3 injected CMS-03 into a **Geosynchronous Transfer Orbit (GTO)**. The satellite then uses its own onboard propulsion system to gradually perform orbit-raising maneuvers to reach its final geostationary altitude of ~35,786 km.
- **Statement D is incorrect:** It is not part of the NavIC navigation system. CMS-03 is a dedicated military communication satellite designed specifically for the Indian Navy. It replaces the aging GSAT-7 (Rukmini) satellite to provide secure, real-time voice, video, and data links between warships, submarines, aircraft, and shore commands across the Indian Ocean Region

Q 69. Ans. D

Explanation:

- **Advanced Medium Combat Aircraft (AMCA):** This is an indigenously developed fifth-generation stealth multirole fighter jet. It is being designed by the Aeronautical Development Agency (ADA) and the Defence Research and Development Organisation (DRDO), with manufacturing planned by Hindustan Aeronautics Limited (HAL).
- **Zorawar light tank:** This is an indigenous light tank designed specifically for high-altitude operations, such as those required in the mountainous terrain of Ladakh. It is jointly developed by the DRDO and private sector defense manufacturer Larsen & Toubro (L&T).
- **Rafale** is a French twin-engine, multirole fighter aircraft designed and built by Dassault Aviation. While India procured these jets for the Indian Air Force, they are not indigenous platforms developed under the "Make in India" initiative.

Q 70. Ans. D

Explanation:

- PM Research Fellowship (PMRF) is not a component of the BIRAC ecosystem. The Prime Minister's

Research Fellows (PMRF) scheme is administered by the Ministry of Education to improve the quality of research in higher educational institutions (like IITs, IISc, IISERs, and central universities) and attract the best talent into research.

- **Biotechnology Ignition Grant (BIG):** This is a flagship scheme of BIRAC. It provides vital early-stage funding (up to ₹50 lakhs) to biotech startups and individual entrepreneurs to help establish proof-of-concept for their ideas.
- **Small Business Innovation Research Initiative (SBIRI):** This is an early-stage, innovation-focused Public-Private Partnership (PPP) initiative operationalised by BIRAC to support discovery and innovation in the biotech industry, particularly for small and medium enterprises.
- **BioNEST (Bioincubators Nurturing Entrepreneurship for Scaling Technologies):** This is a key BIRAC program designed to create and support specialised bio-incubation centers across the country, providing biotech startups with necessary infrastructure, mentorship, and equipment.

Q 71. Ans. D

Explanation:

- The formula for calculating the work done by a constant force is:
 - ✓ $\text{Work} = \text{Force} \times \text{Displacement} \times \cos(\text{angle})$
- Here is what those terms mean for your question:
 - ✓ Force is 25 Newtons.
 - ✓ Displacement is 4 metres.
- The angle is 0 degrees. Because the problem states the object is displaced exactly in the direction of the force, there is no angle difference between the push and the movement.
- The mathematical value for the cosine of 0 degrees is simply 1. This means the entire force is contributing directly to the work.
- Therefore, the calculation is: $\text{Work} = 25 \times 4 \times 1$
 $\text{Work} = 100 \text{ Joules}$

Q 72. Ans. B

Explanation:

- To find the distance covered, we use the standard motion formula:
 - ✓ $\text{Distance} = (\text{Initial Velocity} \times \text{Time}) + \frac{1}{2} \times \text{Acceleration} \times (\text{Time squared})$
- Initial Velocity is 0 (since the car starts from rest).
- Acceleration is 4.
- Time is 5 seconds.

- **The calculation step-by-step:**

- ✓ Distance = $(0 \times 5) + 1/2 \times 4 \times (5 \times 5)$
 - Distance = $0 + 2 \times 25$
 - Distance = 50 meters

Q 73. Ans. C

Explanation:

- **The formula for calculating electrical energy consumption is:**
 - ✓ Energy = Power x Time
- Power is 25 kilowatts (kW).
- Time is 12 hours.
- **The calculation step-by-step:**
 - ✓ Energy = 25×12
 - Energy = 300 kilowatt-hours (kWh)
 - ✓ In commercial electricity billing, 1 unit is exactly equal to 1 kilowatt-hour (kWh).
- Therefore, 300 kWh is equal to 300 units.

Q 74. Ans. B

Explanation:

- The pitch of a sound is directly determined by its frequency.
- Frequency refers to the number of sound waves (vibrations) that pass a fixed point in one second. It is measured in Hertz (Hz).
- A high frequency means the waves are packed closely together, which produces a high-pitched sound (like a whistle or a bird chirping).
- A low frequency means the waves are spread further apart, which produces a low-pitched sound (like a bass drum or thunder).
- Amplitude determines the loudness or volume of the sound, not the pitch. A taller wave means a louder sound.
- Velocity is the speed at which the sound travels through a medium (like air or water).
- Wavelength is the physical distance between two consecutive waves. While it is related to frequency (a shorter wavelength means a higher frequency), pitch is formally defined by the frequency itself.

Q 75. Ans. C

Explanation:

- When a satellite transitions to an orbit of a greater radial distance from a primary planetary body, its mechanical energy components undergo specific alterations governed by orbital mechanics.
- **Increase in Gravitational Potential Energy:**
 - ✓ The gravitational potential energy of a two-body system is intrinsically negative,

approaching zero asymptotically as the radial distance approaches infinity. The relationship is mathematically defined as $U = -GMm/r$, where G is the universal gravitational constant, M and m are the respective masses, and r is the orbital radius. As the orbital radius increases, the absolute magnitude of the fraction decreases. Because the overall value becomes less negative and closer to zero, this constitutes a net increase in gravitational potential energy. In physical terms, work is performed against the gravitational field to elevate the satellite to a higher altitude, thereby increasing its stored energy.

- **Decrease in Kinetic Energy:**

- ✓ For a satellite to maintain a stable circular orbit, the gravitational force must provide the exact requisite centripetal force. By equating the gravitational force to the centripetal requirement, it is established that the orbital velocity is inversely proportional to the square root of the orbital radius. The kinetic energy, expressed mathematically as $K = GMm/2r$, exhibits a direct inverse relationship with the radius. Therefore, as the satellite occupies a higher orbit with a larger radius, its required orbital velocity diminishes, resulting in a corresponding decrease in its kinetic energy.

Q 76. Ans. C

Explanation:

- The principle of conservation of mechanical energy states that the total mechanical energy of an isolated system remains constant in the absence of non-conservative forces like air resistance.
- **Initial Energy State:** At the point of vertical projection (ground level), the potential energy is zero. Therefore, the total mechanical energy is entirely kinetic.
 - ✓ Initial Kinetic Energy = $1/2mv^2$ (where ' m ' is the mass of the body and ' v ' is the initial velocity).
 - ✓ Total Initial Energy = $1/2mv^2$
- **Energy State at Height ' h ':** As the body ascends, kinetic energy is converted into gravitational potential energy. At a specific height ' h ', the problem establishes the condition that the kinetic energy is exactly equal to the potential energy.
 - ✓ Gravitational Potential Energy at height ' h ' = mgh (where ' g ' is the acceleration due to gravity).

- Because Kinetic Energy = Potential Energy at this height, we can express the total energy at this point purely in terms of potential energy:

- ✓ Total Energy at height 'h' = Potential Energy + Kinetic Energy

- Total Energy at height 'h' = Potential Energy + Kinetic Energy

- ✓ Total Energy at height 'h' = $2(mgh)$

- Equating the States: By the law of conservation of energy, the total initial energy must equal the total energy at height 'h'.

- ✓ $\frac{1}{2}mv^2 = 2mgh$

- To isolate 'h', divide both sides by (2mg):

- ✓ $h = (1/2mv^2)/(2mg)$

- $h = v^2/4g$

Q 77. Ans. B

Explanation:

- A virtual image with a magnification greater than one means the image is upright, cannot be projected on a screen, and is physically larger than the original object.
 - ✓ Convex mirrors always form virtual images that are smaller than the object (magnification is less than 1). This is why they are used as passenger-side car mirrors to show a wider area.
 - ✓ Concave lenses also always form virtual images that are smaller than the object.
 - ✓ Plane mirrors form virtual images that are the exact same size as the object (magnification is exactly 1).
 - ✓ Concave mirrors will form a virtual, upright, and magnified image (magnification greater than 1) when the object is placed very close to the mirror (specifically, between the mirror's pole and its focal point). This is the principle behind makeup and shaving mirrors.

Q 78. Ans. C

Explanation:

- Consider a battery of emf E and internal resistance r connected to an external resistance R . The current in the circuit is:
 - ✓ $I = E / (R + r)$
- The power delivered to the external resistor R is:
 - ✓ $P = I^2R = E^2R / (R + r)^2$
- To find the value of R for which P is maximum, we differentiate P with respect to R and set $dP/dR = 0$. Carrying out this differentiation gives the condition:
 - ✓ $R = r$

- That is, the power delivered to the external resistor is maximum when the external resistance equals the internal resistance of the battery. At this point, the maximum power transferred is $E^2/4r$, and the efficiency of transfer is exactly 50%, since half the total power is dissipated inside the battery and half in the external resistor.

- ✓ **Equal to zero:** This is the short-circuit condition. The current is maximum, but since $R = 0$, the power delivered to the external resistor ($P = I^2R$) is zero. All the power is dissipated internally.

- ✓ **Equal to battery emf:** This is dimensionally meaningless. Emf is measured in volts and resistance in ohms, so an external resistance cannot be "equal to" the emf. This is a conceptual distractor.

- ✓ **Infinite:** This is the open-circuit condition. The current becomes zero, so the power delivered ($P = I^2R$) is again zero.

- Therefore, maximum power is supplied to the external resistor when $R = r$.

Q 79. Ans. D

Explanation:

- The efficiency of an electrical cell is defined as the ratio of the useful output power delivered to the external circuit to the total power generated by the cell.
- Let R represent the external resistance and r represent the internal resistance of the cell.
- **The formula for efficiency is:** Efficiency = R divided by $(R + r)$
- **To find the condition where the efficiency is exactly zero, we set the formula to zero:** $0 = R$ divided by $(R + r)$
- For any fraction to equal zero, its numerator must be zero. Therefore, the external resistance R must equal zero.
- In physical terms, an external resistance of zero creates a short circuit. The cell outputs maximum current, but all of the electrical energy is consumed by the cell's own internal resistance, dissipating as heat. Because no useful power is delivered to an external load, the efficiency of the cell drops completely to zero.

Q 80. Ans. B

Explanation:

- **RCBO (Residual Current Breaker with Overcurrent):** This device serves a dual purpose. It combines the functions of an MCB (Miniature

Circuit Breaker) and an RCD (Residual Current Device) into one single unit. It monitors the circuit for earth leakage to protect people from electric shocks, and it also monitors overcurrent and short circuits to protect the wiring from overheating and catching fire.

- **ELCB (Earth Leakage Circuit Breaker):** This is an older type of safety device designed strictly to detect earth leaks. By itself, it does not contain the internal mechanisms required to trip the circuit in the event of an overcurrent or a short circuit.
- **Ammeter:** This is merely a measuring instrument used to display the amount of electrical current flowing through a circuit. It offers absolutely no automatic protection.
- **Isolator:** This is a mechanical switch used to safely disconnect a circuit for maintenance. While it can carry the normal load current, it does not automatically detect faults or trip the circuit on its own.

Q 81. Ans. B

Explanation:

- Hydrochloric acid (HCl) is a strong acid, and sodium hydroxide (NaOH) is a strong base. When they react together, they produce sodium chloride (NaCl) and water.
- Since, equal volumes of both solutions are mixed, and both have the exact same concentration (0.1 M).
- Because the concentration and volume are identical, the number of acid molecules perfectly matches the number of base molecules. They completely neutralise each other, leaving no excess acid or base behind.
- The resulting solution is simply salt water. Sodium chloride is a neutral salt, meaning it does not alter the hydrogen ion concentration of the water.
- Therefore, the final solution is perfectly neutral, which always corresponds to a pH of exactly 7 at standard room temperature.

Q 82. Ans. A

Explanation:

- **Europium** is a rare earth metal that is highly valued in materials science for its unique luminescent properties. When europium is oxidised to its plus-three state, it produces a brilliant, highly efficient red light when excited by ultraviolet radiation or electrons.
 - ✓ This specific optical property makes it the

foundational element for red phosphors. It is heavily utilized in manufacturing modern light-emitting diode (LED) screens, fluorescent lamps, and legacy cathode ray tube (CRT) televisions to produce the red pixels necessary for full-color displays.

- **Scandium:** This rare earth element is primarily alloyed with aluminum to create lightweight, high-strength materials for aerospace components and high-end sports equipment. It is not used as a phosphor.
- **Hafnium:** This transition metal is an excellent absorber of neutrons and is highly resistant to corrosion. Its primary application is in the manufacturing of control rods for nuclear reactors.
- **Zirconium:** Similar to hafnium, zirconium is highly resistant to heat and corrosion. It is widely used to make the outer cladding for nuclear fuel rods and to line pipes in chemical processing plants.

Q 83. Ans. A

Explanation:

- **Indium tin oxide** is widely recognized as the industry standard for transparent conducting electrodes. It possesses a rare and highly valuable combination of two physical properties: it conducts electricity very well, and it is highly transparent to visible light.
- Because of this dual capability, it can be applied as a thin, invisible film over glass or plastic. This allows devices to register electrical inputs from a finger touch while still allowing the underlying visual display to be seen clearly, making it essential for modern touchscreens, solar cells, and smart window applications.
- Silicon carbide is a robust semiconductor primarily used in high-voltage power electronics and electric vehicle components.
- Titanium dioxide is most commonly used as a bright white pigment in paints, sunscreens, or as a photocatalyst, not as a transparent conductor.
- Boron nitride is an excellent thermal conductor but acts as an electrical insulator, making it fundamentally incompatible for use as an electrode.

Q 84. Ans. D

Explanation:

- Carboplatin is a widely used chemotherapy medication that belongs to a specific class of drugs known as platinum-based antineoplastic agents.
- The core molecular structure of carboplatin contains

a central platinum atom. When administered, this heavy metal complex enters cancer cells and binds directly to their DNA. The platinum atom creates structural cross-links in the DNA strands, which effectively blocks the cells from copying their genetic material. Because the cancer cells can no longer divide and multiply, they ultimately die.

- Carboplatin was specifically developed as a second-generation alternative to the original platinum drug, cisplatin. While it uses the exact same platinum-based mechanism to destroy tumors, its modified chemical structure makes it much less toxic to the patient's kidneys and nervous system.

Q 85. Ans. A

Explanation:

- **Ulipristal acetate** is classified pharmacologically as a selective progesterone receptor modulator. It functions by binding directly to the human progesterone receptor, thereby preventing the natural hormone progesterone from occupying its normal binding sites. Depending on the timing of administration during the menstrual cycle, this mechanism effectively delays or completely inhibits the release of an egg from the ovary (ovulation), making it a highly effective agent for emergency contraception.
- **Metformin:** This is an oral antidiabetic medication used primarily to manage type 2 diabetes by decreasing hepatic glucose production and improving insulin sensitivity.
- **Cetirizine:** This is a second-generation antihistamine utilised to treat allergy symptoms by blocking histamine receptors.
- **Omeprazole:** This is a proton pump inhibitor designed to reduce stomach acid secretion, commonly prescribed for gastroesophageal reflux disease and peptic ulcers.

Q 86. Ans. A

Explanation:

- Honey is a naturally acidic substance, typically possessing a pH level that ranges between 3.4 and 6.1. This inherent acidity is primarily due to the presence of various organic acids, most notably gluconic acid, which is produced during the enzymatic breakdown of sugars by the bees.
- When honey is placed in an ordinary, unlined iron container, a chemical reaction occurs. The organic acids within the honey actively corrode the elemental iron surface.

- This reaction leaches iron ions directly into the honey. As a result, the honey undergoes undesirable chemical changes, typically developing a dark, unappealing color and a sharp metallic taste that ruins its quality and renders it unsuitable for consumption.
- To prevent this chemical degradation and maintain the purity of the product, honey must always be stored in containers made of chemically inert materials. This is why commercial packaging relies on glass, food-grade plastics, or specially treated stainless steel that is highly resistant to acidic corrosion.

Q 87. Ans. D

Explanation:

- Reforming, specifically catalytic reforming, is a major chemical process used in petroleum refineries. Its primary purpose is to take low octane petroleum fractions, typically straight chain hydrocarbons, and convert them into high octane liquid products.
- During this process, the straight chain molecules are subjected to high heat and pressure in the presence of a catalyst, such as platinum. This causes the molecules to rearrange their structures. The straight chains fold and bond into branched chain hydrocarbons or aromatic rings.
- Because branched and aromatic hydrocarbons burn much more smoothly and predictably in an internal combustion engine than straight chain hydrocarbons, this structural rearrangement significantly increases the overall octane rating of the fuel.
- Fractional distillation is a physical separation process that divides crude oil into different parts based on their boiling points. It only separates the existing molecules; it does not alter their internal chemical structures.
- Desalting is a preparatory step that removes inorganic salts from the crude oil before it is processed. This is done to prevent corrosion and scale buildup in the refinery equipment.
- Dewaxing is a process used to remove heavy, solid waxes from lubricating oils. This ensures the final oil product continues to flow properly and does not freeze or thicken excessively at cold temperatures.

Q 88. Ans. C

Explanation:

- PTFE stands for Polytetrafluoroethylene. It is a synthetic fluoropolymer formed by the polymerisation of the monomer tetrafluoroethylene (C_2F_4). It is best known by the commercial brand name Teflon (developed by DuPont).

- **PTFE is widely used as a non-stick coating on cookware because of its exceptional properties:** it is highly chemically inert, resistant to heat, and has an extremely low coefficient of friction, which gives surfaces their characteristic “non-stick” quality. The strength of the carbon-fluorine bonds in its structure is responsible for its chemical stability and non-reactive nature.
- **IgA:** While this antibody is highly concentrated in colostrum and breast milk to provide localized mucosal immunity to the infant’s gastrointestinal tract after birth, it lacks the specific structural domains required to bind to the placental transport receptors during gestation.

Q 89. Ans. C

Explanation:

- **Endoparasitism:** In this type of parasitic relationship, the parasite lives inside the internal organs or tissues of its host. The tapeworm resides internally within the human digestive tract, absorbing the host’s digested nutrients and causing harm or malnutrition to the host.
- **Ectoparasitism:** This occurs when a parasite lives on the outside surface of the host’s body rather than inside it. Common examples include ticks, fleas, leeches, and lice.
- **Mutualism:** This is a symbiotic relationship where both participating species benefit from the interaction and often cannot survive without each other. Because the tapeworm benefits at the direct expense of the human, this is not mutualism.
- **Protocooperation:** This is a mutually beneficial interaction between two species that is helpful but not biologically essential for the survival of either species (such as certain birds that eat parasites off the skin of large mammals). Since the human host is harmed, this does not apply.

Q 90. Ans. C

Explanation:

- Immunoglobulin G (IgG) is the only class of antibody that can successfully cross the human placental barrier during normal pregnancy.
- This unique capability is facilitated by specific transport proteins, known as neonatal Fc receptors, which are expressed on the cells of the placenta. These specialised receptors actively bind to the IgG molecules present in the maternal bloodstream and transport them safely across the tissue layers directly into the fetal circulation. This biological mechanism is essential for fetal survival, as it provides the developing infant with crucial passive immunity, protecting the newborn against a wide array of infectious pathogens during its first few months of life while its own adaptive immune system is still maturing.

- **IgD:** This immunoglobulin functions almost exclusively as an antigen receptor bound to the surface membrane of immature B lymphocytes. It is secreted into the blood plasma in only minute trace amounts and plays no role in maternal-fetal immune transfer.
- **IgM:** As noted in previous principles, the IgM antibody is constructed as a massive pentameric complex. Its exceptionally large molecular weight and physical size completely prevent it from physically passing through the placental filtration barrier.

Q 91. Ans. A

Explanation:

- **The extracellular matrix of animal connective tissue is a complex structural network that provides physical support and biochemical signaling to surrounding cells. It is primarily composed of two main components:**
 - ✓ **Fibrous Proteins:** Collagen provides high tensile strength and structural integrity, while elastin allows tissues (like skin and blood vessels) to stretch and safely recoil to their original shape.
 - ✓ **Ground Substance:** This is a highly hydrated, gel-like material that fills the spaces between the cells and fibers. It is made mostly of proteoglycans and glycoproteins, which trap water to resist physical compression and facilitate the transport of nutrients and cellular waste.
- **Cellulose and pectin:** These are the primary structural components of plant cell walls, not animal tissues.
- **Peptidoglycan and teichoic acids:** These specific molecules form the rigid cell walls of bacteria.
- **Chitin:** This is a tough carbohydrate polymer that makes up the exoskeletons of insects and crustaceans, as well as the cell walls of fungi.

Q 92. Ans. C

Explanation:

- **The genetic code is described as degenerate (or redundant), meaning most amino acids are encoded by more than one codon. Of the 20 standard amino acids, only two are coded by a single codon each: methionine (AUG) and tryptophan (UGG).**

- **Leucine:** Encoded by six codons (UUA, UUG, CUU, CUC, CUA, CUG), one of the most degenerate amino acids in the code.
- **Serine:** Encoded by six codons (UCU, UCC, UCA, UCG, AGU, AGC).
- **Arginine:** Encoded by six codons (CGU, CGC, CGA, CGG, AGA, AGG).

Q 93. Ans. C

Explanation:

- According to the cohesion tension theory, the primary mechanism driving the upward movement of water, or sap, in tall trees is transpiration pull. Water evaporates from the small pores, called stomata, located on the surface of the leaves. This continuous loss of water creates a state of negative pressure, or tension, within the leaf tissues.
- **Cohesion:** Water molecules possess a strong mutual attraction due to hydrogen bonding. This property, known as cohesion, allows the water molecules to form an unbroken, continuous column within the xylem vessels running from the roots all the way to the leaves.
- **Adhesion:** The water molecules also physically adhere to the structural walls of the xylem cells. This adhesion helps support the water column against the downward pull of gravity.
- Because the water molecules are tightly linked together by cohesion, the negative pressure generated by evaporation at the top of the tree mechanically pulls the entire water column upwards, functioning much like liquid being drawn up through a straw.

Q 94. Ans. C

Explanation:

- Adolf Engler and Karl Prantl are the botanists credited with the phylogenetic (evolutionary) system of plant classification. The two German botanists published their monumental work "Die Natürlichen Pflanzenfamilien" (1887–1915) and Engler's "Syllabus der Pflanzenfamilien" (1892), classifying the entire plant kingdom on the basis of evolutionary relationships and arranging families in ascending order of floral complexity.
- Carolus Linnaeus gave the artificial system of classification, based mainly on a single character (the number and arrangement of stamens), not on natural or evolutionary relationships.
- Bentham and Hooker gave the most well-known natural system of classification (in "Genera

Plantarum"), based on overall natural affinities and observable characters, but it was not phylogenetic. It was, in fact, pre-Darwinian in spirit.

- Theophrastus, the "Father of Botany," gave one of the earliest classifications (in "Historia Plantarum"), grouping plants by habit into trees, shrubs, undershrubs, and herbs, which was purely artificial

Q 95. Ans. C

Explanation:

- A lichen is a classic example of a symbiotic (mutualistic) association between a fungus (the mycobiont) and an alga or cyanobacterium (the phycobiont). The two partners exchange complementary benefits.
- **What the alga gets from the fungus:** The fungal partner provides the alga with water, shelter (protection and a physical structure to live in), and mineral nutrients absorbed from the surroundings.

Q 96. Ans. C

Explanation:

- The Government of India Act of 1858, enacted after the revolt of 1857, abolished the East India Company and transferred the powers of government to the Crown. It created the office of the Secretary of State for India, a member of the British cabinet assisted by a fifteen member Council of India, and ended the system of double government by abolishing the Board of Control and the Court of Directors.
- The Indian Councils Act of 1861 introduced the portfolio system, under which each member of the Viceroy's council was placed in charge of a department, and it began the process of associating Indians with law making by allowing the Viceroy to nominate non official members.
- The Act of 1858 also changed the designation of the Governor General to Viceroy, Lord Canning being the first Viceroy. The Act of 1861 initiated the decentralisation of legislative power by restoring it to the Bombay and Madras presidencies and empowered the Viceroy to issue ordinances during an emergency, laying the foundations of representative institutions in India.

Q 97. Ans. A

Explanation:

- **Statement 1 is correct:** Critics described the Constitution as a borrowed document and a carbon copy of the Government of India Act of 1935,

since a large number of its provisions were drawn substantially from that Act, and some members of the Constituent Assembly called it a bag of borrowings.

- **Statement 2 is incorrect:** On the contrary, the Constitution was criticised as un-Gandhian, because in its original form it did not give a central place to village panchayats and ignored the Gandhian vision of decentralised village republics; one member, K. Hanumanthaiya, complained that the Assembly had produced an elephant when the people wanted the music of a veena.
- Other criticisms were that the Constitution was a paradise of lawyers because of its legalistic and complex language, that it was unwieldy and elephantine in size, that it borrowed without acknowledging Indian conditions, and that it diluted federalism with a strong unitary bias. Defenders such as B. R. Ambedkar replied that borrowing tested provisions was not plagiarism and that the document suited the needs and conditions of the country.

Q 98. Ans. A

Explanation:

- **Statement 1 is correct:** India provides for a single citizenship for the entire country, so there is no separate citizenship of the states, and all citizens owe allegiance to the Union alone and enjoy uniform rights across the territory, subject to a few special provisions for certain states.
- **Statement 2 is correct:** An Overseas Citizen of India is a foreign national of Indian origin and is not a full citizen of India; the scheme, introduced by the Citizenship Amendment Act of 2005, grants certain facilities such as a multiple entry visa and parity with non-resident Indians in some matters, but it does not confer citizenship.
- **Statement 3 is incorrect:** An Overseas Citizen of India does not enjoy political rights and therefore cannot vote, cannot be a member of a legislature, cannot hold constitutional offices such as President, Vice-President, or judge of the higher judiciary, and cannot normally be appointed to public services. India follows single citizenship on the British model, in contrast to the dual citizenship found in federations such as the United States.

Q 99. Ans. A

Explanation:

- **Article 29(2) explicitly states:** “No citizen shall be denied admission into any educational institution

maintained by the State or receiving aid out of State funds on grounds only of religion, race, caste, language or any of them.”

- A crucial point to note for civil services preparation is that while Article 29 is placed under the heading of “Cultural and Educational Rights” of minorities, Article 29(2) is an individual right guaranteed to every citizen, regardless of whether they belong to a majority or minority community (as upheld by the Supreme Court in the landmark *State of Madras v. Champakam Dorairajan* case).

Q 100. Ans. A

Explanation:

- **Statement 1 is correct:** The Directive Principles are non-justiciable, meaning they are not legally enforceable by any court, and their backing is moral and political rather than legal; nonetheless Article 37 declares them fundamental in the governance of the country and a duty of the State to apply in making laws.
- **Statement 2 is correct:** Article 350A directs every State and local authority to provide adequate facilities for instruction in the mother tongue at the primary stage of education to children belonging to linguistic minority groups, and it is a directive located outside Part IV.
- **Statement 3 is incorrect:** There are several directives outside Part IV, such as Article 335 on the claims of the scheduled castes and tribes to services and posts, Article 350A on instruction in the mother tongue, and Article 351 on the development of the Hindi language; although placed elsewhere, they are non-justiciable and not enforceable by the courts, in the same manner as the directives within Part IV, and the judiciary has held that they are to be given importance equivalent to the directives in Part IV since all parts of the Constitution must be read together.

Q 101. Ans. C

Explanation:

- **Assertion is true:** The basic structure doctrine, propounded in the *Kesavananda Bharati* case of 1973, applies to amendments of the Constitution made under Article 368, and an amendment that damages or destroys the basic structure can be struck down even though Parliament otherwise has wide amending power.
- **Reason is false:** Ordinary laws made by Parliament are tested for their conformity with the provisions

of the Constitution, especially the fundamental rights, and not against the basic structure doctrine as such, which was evolved specifically to control the amending power. The validity of an ordinary law is judged by whether it violates a specific constitutional provision.

- Elements held to form part of the basic structure include the supremacy of the Constitution, the sovereign, democratic, and republican character of the polity, the secular character, the separation of powers, judicial review, the rule of law, free and fair elections, the independence of the judiciary, and the balance between fundamental rights and directive principles. The doctrine has thus become the principal judicial check on the constituent power of Parliament.

Q 102. Ans. C

Explanation:

- **Statement 1 is correct:** The Council of Ministers, which may number sixty to seventy ministers, rarely meets as a single body to transact government business; it is a body whose collective responsibility is enforced through the Lok Sabha, but the actual exercise of executive power is concentrated in its inner core.
- **Statement 2 is correct:** The Cabinet is a much smaller body of fifteen to twenty senior most ministers within the Council of Ministers, and it is the real centre of authority that directs the executive government, formulates policy, coordinates departments, and meets frequently to take decisions.
- **Statement 3 is incorrect:** The kitchen cabinet is an informal and extra constitutional body, consisting of the Prime Minister and a few influential colleagues or friends and advisers on whom the Prime Minister relies for advice; it is not recognised by the Constitution and is criticised for bypassing the cabinet. The Council of Ministers consists of three categories, namely cabinet ministers, ministers of state, and deputy ministers, and the 91st Amendment limits its total size to fifteen per cent of the strength of the Lok Sabha.

Q 103. Ans. A

Explanation:

- Article 350B provides for a Special Officer for Linguistic Minorities, whose duty is to investigate all matters relating to the safeguards provided for linguistic minorities under the Constitution and to

report to the President upon those matters at such intervals as the President may direct.

- The Special Officer, designated the Commissioner for Linguistic Minorities, is appointed by the President, not by the Chief Justice of India. The reports of the Commissioner are placed before each House of Parliament and sent to the governments of the states concerned.
- Article 350B was inserted by the 7th Constitutional Amendment Act of 1956 on the recommendation of the States Reorganisation Commission, since the reorganisation of states on a linguistic basis raised concerns about the protection of linguistic minorities. The office functions under the administrative control of the Union ministry dealing with minority affairs and works to safeguard the linguistic rights of minorities across the country.

Q 104. Ans. A

Explanation:

- Judicial restraint is the principle that courts should avoid interfering with the actions of the legislature and executive unless there is a clear constitutional violation. It emphasizes judicial self-discipline and respect for the separation of powers. In contrast, judicial activism involves a more proactive role for courts in interpreting the Constitution and addressing public issues, while judicial review is the power to examine the constitutionality of laws and executive actions. Judicial overreach refers to situations where courts are perceived to exceed their constitutional mandate.
- Judicial activism in India was promoted by judges such as V. R. Krishna Iyer and P. N. Bhagwati from the late 1970s. While it has expanded access to justice and protected fundamental rights, critics warn that excessive activism may amount to judicial adventurism and may upset the separation of powers by encroaching upon the domains of the legislature and the executive.

Q 105. Ans. C

Explanation:

- **Statement 1 is correct:** Under Article 200 the Governor may, instead of giving or withholding assent, reserve a bill passed by the state legislature for the consideration of the President, and reservation is obligatory where the bill would derogate from the powers of the High Court.
- **Statement 2 is incorrect:** The pardoning power of the Governor under Article 161 does not extend to a

death sentence and does not extend to punishments inflicted by a court martial, whereas the pardoning power of the President under Article 72 extends to both. The Governor can, however, suspend, remit, or commute a death sentence.

- **Statement 3 is correct:** The Governor enjoys constitutional discretion, such as reserving a bill for the President or recommending President's rule, and situational discretion, such as appointing a Chief Minister when no party has a clear majority or seeking a vote of confidence; under Article 163 the Council of Ministers aids and advises the Governor except where he is required to act in his discretion.

Q 106. Ans. B

Explanation:

- **Statement 1 is correct:** The National Legal Services Authority was set up under the Legal Services Authorities Act of 1987 to provide free and competent legal services to the weaker sections and to organise Lok Adalats for amicable settlement of disputes.
- **Statement 2 is correct:** Family Courts were established under the Family Courts Act of 1984 to promote conciliation and secure speedy settlement of disputes relating to marriage and family affairs.
- **Statement 3 is incorrect:** Gram Nyayalayas, established under the Gram Nyayalayas Act of 2008, are not High Court level bodies; they are the lowest tier of the judiciary, set up at the intermediate panchayat level to provide access to justice at the grassroots and to try certain civil and criminal cases.
- **Statement 4 is correct:** Permanent Lok Adalats, provided for by a 2002 amendment to the Legal Services Authorities Act for disputes concerning public utility services, have adjudicatory powers and can decide a dispute on merits even where the parties fail to reach a settlement, up to a prescribed pecuniary limit; an ordinary Lok Adalat award is final, is deemed a civil court decree, and admits of no appeal.

Q 107. Ans. A

Explanation:

- Article 343 declares Hindi in the Devanagari script to be the official language of the Union, while permitting the continued use of English; the Official Languages Act of 1963 allows the indefinite continuation of English alongside Hindi. Article 345 allows a state to adopt one or more languages in use in the state as its official language, and Article 348 deals with the language to be used in the Supreme Court and the High Courts and in the texts of laws.

- Tamil was the first language to be accorded classical language status, in 2004, and the criteria for the status include high antiquity of early texts, a body of ancient literature regarded as a valuable heritage, and an original literary tradition not borrowed from another speech community; Sanskrit, Telugu, Kannada, Malayalam, and Odia were among the languages later accorded the status.

Q 108. Ans. C

Explanation:

- **Statement 1 is correct:** Under Article 299 all contracts made in the exercise of the executive power of the Union or a state must be expressed to be made by the President or the Governor, and must be executed on their behalf by persons authorised for the purpose; a contract not so made is not enforceable against the government.
- **Statement 2 is incorrect:** Under Article 361 the President and the Governors are not answerable to any court for the exercise and performance of the powers and duties of their office, no criminal proceedings can be instituted against them during their term, and no process for their arrest or imprisonment can issue from any court during their term.
- **Statement 3 is correct:** Under Article 300 the Government of India may sue or be sued in the name of the Union of India, and the government of a state in the name of the state, in relation to their respective affairs, in the same manner as an ordinary person, subject to any law made by Parliament or the state legislatures. The property of the Union and the states is governed by Articles 294 to 296.

Q 109. Ans. C

Explanation:

- **Statement 1 is correct:** The None of the Above option, which allows a voter to reject all the contesting candidates, was introduced on the electronic voting machines following a Supreme Court judgment of 2013, and it is intended to give expression to the voter's right to disapproval and to secrecy of the choice not to vote for anyone.
- **Statement 2 is incorrect:** The Model Code of Conduct does not have statutory backing under an Act of Parliament; it is a set of norms evolved with the consensus of political parties and enforced by the Election Commission, coming into force from the date the election schedule is announced and remaining in force till the completion of the process.

- **Statement 3 is correct:** Article 82 provides for the readjustment of the allocation of seats in the Lok Sabha and the division of states into territorial constituencies after the completion of each census, through a Delimitation Commission, though the allocation has been frozen on the basis of the 1971 census until the first census after 2026 under amendments to the Constitution.

Q 110. Ans. D

Explanation:

- The National Integration Council is an extra constitutional advisory body, chaired by the Prime Minister, that brings together ministers, chief ministers, leaders of political parties, and eminent persons to examine and address problems such as communalism, casteism, regionalism, and linguism that threaten national integration. Devolution is NOT a primary concern of the National Integration Council
- The National Foundation for Communal Harmony is an autonomous organisation that works under the Ministry of Home Affairs, and it promotes communal harmony and national integration, and provides assistance for the physical and psychological rehabilitation of children who are rendered orphan or destitute by communal, caste, ethnic, or terrorist violence.
- National integration is the awareness of a common identity among the citizens that transcends differences of religion, region, language, and caste, while preserving the country's diversity. Obstacles to it include communalism, regionalism, casteism, and linguistic fanaticism, and the National Integration Council was first set up in 1961 following a conference convened to find ways of combating these divisive tendencies.

Q 111. Ans. A

Explanation:

- **Statement 1 is correct:** Germany won the FIH Men's Junior Hockey World Cup 2025 by defeating Spain in the final to claim their record-extending eighth title. The final ended 1–1 in regulation time, after which Germany prevailed 3–2 in the shoot-out. This marked Germany's 10th appearance in the final and their eighth championship victory.
- **Statement 2 is incorrect:** After suffering a 5–1 defeat to Germany in the semi-finals, India competed in the third-place playoff against Argentina. The Indian team defeated Argentina 4–2, thereby securing the

Bronze Medal. This was India's first-ever bronze medal in the Men's Junior Hockey World Cup and its first medal in the tournament since winning the title in 2016.

Q 112. Ans. B

Explanation:

- Vivek Chaturvedi was appointed as the Chairman of the Central Board of Indirect Taxes and Customs (CBIC).
- Raj Kumar Goyal was appointed as the Chief Information Commissioner (CIC).
- Nitin Nabin was appointed as the National President of the Bharatiya Janata Party (BJP).
- Praveen Kumar was given additional charge as Director General (DG), Border Security Force (BSF).

Q 113. Ans. A

Explanation:

- Fátima Bosch of Mexico won the Miss Universe 2025 title at the 74th Miss Universe pageant held in Thailand. She succeeded Victoria Kjær Theilvig of Denmark and became the fourth Mexican to win the crown.

Q 114. Ans. D

Explanation:

- The UNESCO-listed Ningaloo Reef is located off the coast of Western Australia, in Australia.
- The reef forms part of the Ningaloo Coast, which was inscribed on the UNESCO World Heritage List in 2011.
- It is one of the world's largest fringing coral reefs, stretching for about 260 km along Australia's northwest coast.
- The reef is renowned for its rich marine biodiversity, including whale sharks, manta rays, sea turtles, dugongs, and hundreds of coral species.
- A recent severe marine heatwave caused extensive coral bleaching, with reports indicating that nearly 70% of corals were affected or lost in some surveyed areas, highlighting the growing impact of climate change on coral reef ecosystems.

Q 115. Ans. C

Explanation:

- **Statement 1 is correct:** The Kimberley Process (KP) Plenary selected India to assume the Chairpersonship of the Kimberley Process. Before formally taking over as Chair, India will serve as the Vice Chair from 25 December 2025, in accordance with the KP's succession mechanism.

- **Statement 2 is correct:** India's assumption of the Chairpersonship in 2026 marks the third time that the country has been entrusted with leading the Kimberley Process. This reflects India's significant role in the global diamond trade and the confidence reposed in it by the international community.

Q 116. Ans. B

Explanation:

- Scientists from the Department of Genetics and Plant Breeding at Banaras Hindu University (BHU) spent over 14 years using mutagenesis (a process that induces genetic mutations to improve crop traits) to successfully revive this heritage variety
- The BHU successfully shortened the plant height to about 105 cm from 165 cm, reduced the maturity period to 120 days, and boosted the yield by 50–70% while completely retaining its unique sugar-crystal appearance and signature aroma. The state government has since branded it as “Vindhya Black Rice.”

Q 117. Ans. B

Explanation:

- **Yang Shuang-zi** won the International Booker Prize in 2026 for the novel *Taiwan Travelogue*, translated into English by Lin King. This achievement marked a historic milestone, as it was the first book translated from Mandarin Chinese (specifically Taiwanese Mandarin) to win this prestigious literary award.

Q 118. Ans. C

Explanation:

- Prime Minister Narendra Modi recently unveiled a 77-foot bronze statue of Lord Ram during the 550th-year celebrations (Sardha Panchashatamanotsava) of the Shree Samsthan Gokarn Partagali Jeevottam Math in Canacona, South Goa. He also inaugurated a Ramayana Theme Park, and released a commemorative coin and special postal stamp on the occasion.
- The Shree Samsthan Gokarn Partagali Jeevottam Math is the first Gowd Saraswat Brahmin Vaishnava Math and is headquartered at Partagali in South Goa, on the banks of the Kushavati River.

Q 119. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Global Environment Outlook (GEO) is the flagship environmental assessment report of the United Nations

Environment Programme (UNEP), whose headquarters is located in Nairobi, Kenya. However, it is released by UNEP, not by the United Nations Environment Assembly (UNEA).

- **Statement 2 is correct:** The theme/title of the seventh edition of the Global Environment Outlook (GEO-7) is “A Future We Choose: The report states that strategic investments in climate stability, biodiversity, and pollution reduction could yield USD 20 trillion annually by 2070, whereas inaction risks devastating both economies and ecosystems.

Q 120. Ans. A

Explanation:

- Sarita Yolmo recently created history by becoming the first woman Travelling Ticket Examiner (TTE) to be assigned the prestigious “Tour Duty” on the Darjeeling Himalayan Railway (DHR), popularly known as the Toy Train. This marks the first such appointment in the railway's nearly 145-year history.
- She is a veteran employee of the Northeast Frontier Railway (NFR) and is based at New Jalpaiguri (NJP).
- The Darjeeling Himalayan Railway is a UNESCO World Heritage Site and one of India's most iconic mountain railways.

Q 121. Ans. C

Explanation:

- The Kumbhalgarh Wildlife Sanctuary, recently declared its Eco-Sensitive Zone (ESZ), is located in the Aravalli Range of Rajasthan.
- The sanctuary surrounds the famous Kumbhalgarh Fort and extends across the districts of Rajsamand, Udaipur, and Pali.
- It is known for its rugged Aravalli terrain and rich biodiversity, including leopard, sloth bear, wolf, hyena, and jungle cat.
- The sanctuary is one of the few protected areas in Rajasthan where the Indian wolf is regularly found.

Q 122. Ans. B

Explanation:

- Nnena Kalu recently became the first artist with a learning disability to win the prestigious Turner Prize.
- She was recognised for her distinctive large-scale sculptural works created using layered materials such as paper, tape, fabric, and found objects.
- Her work has been widely praised for its originality, expressive power, and innovative approach to contemporary art.

Q 123. Ans. D

Explanation:

- **Pair 1 is correctly matched:** Raveena Tandon was named PETA India's Person of the Year 2025 in recognition of her advocacy for animal welfare, wildlife protection, and compassionate living.
- **Pair 2 is correctly matched:** Supriya Sahu, Additional Chief Secretary of Tamil Nadu, received the UNEP Champions of the Earth Award 2025, the United Nations' highest environmental honour, under the "Inspiration and Action" category.
- **Pair 3 is correctly matched:** Donald Trump was awarded the inaugural FIFA Peace Prize – Football Unites the World by FIFA President Gianni Infantino during the 2026 FIFA World Cup Final Draw ceremony.

Q 124. Ans. B

Explanation:

- Diya Binu Pullikkakandam recently became India's youngest municipal chairperson and the first representative from Generation Z (Gen Z) to hold the post. At the age of 21, she was elected Chairperson of Pala Municipality in Kottayam district, Kerala.
- She won the municipal election as an independent candidate from Pala Municipality.
- Her elevation marked a significant generational shift in local governance, making her the country's youngest municipal chairperson.

Q 125. Ans. B

Explanation:

- Gujarat has officially reclaimed its status as a tiger-present state. The National Tiger Conservation Authority (NTCA) confirmed the permanent residency of a male tiger in the Ratanmahal Wildlife Sanctuary (located in Dahod district), clearing the way for Gujarat's inclusion in the All India Tiger Census.
- This development makes Gujarat the only state in India where three apex big cats, the Asiatic lion, leopard, and tiger coexist simultaneously in the wild.
- **Additional Facts:**
 - ✓ Ratanmahal Wildlife Sanctuary – Known primarily for its population of sloth bears.
 - ✓ Jessore Sloth Bear Sanctuary – Established mainly for the conservation of sloth bears.
 - ✓ Purna Wildlife Sanctuary – Located in the Dangs region and known for its rich biodiversity, but not the site that restored Gujarat's Tiger State status.

Q 126. Ans. B

Explanation:

- Punjab inaugurated its first-ever dog sanctuary as a pilot initiative to address the growing stray dog population in a humane and systematic manner.
- The sanctuary has been established at the Haibowal Dairy Complex in Ludhiana, with a capacity to house around 500 dogs.
- It aims to provide shelter, medical care, sterilisation, and treatment for stray dogs, while reducing dog-bite incidents in the city.
- The initiative follows Supreme Court guidelines on stray dog management, and based on its success, more such sanctuaries are planned across Punjab.

Q 127. Ans. C

Explanation:

- Carlos Alcaraz won the 2026 Australian Open men's singles title. By securing this victory, he became the youngest man in the Open Era to complete a career Grand Slam. In the championship match, he defeated the ten-time Australian Open champion Novak Djokovic.

Q 128. Ans. D

Explanation:

- Karnataka topped the medal tally at the second edition of the Khelo India Beach Games 2026 (held 5–10 January 2026 at Ghoghla Beach, Diu), with 3 gold, 2 silver, and 6 bronze. Tamil Nadu finished second and Manipur third. The mascot of the Games was "Pearl," a dolphin, and the event was organised by the Union Territory of Dadra and Nagar Haveli and Daman and Diu under the technical supervision of the Sports Authority of India.

Q 129. Ans. B

Explanation:

- Bharat-VISTAAR is a multilingual, AI-based digital platform that integrates agricultural services and provides customised advisory support to farmers, announced in the Union Budget 2026–27 and rolled out in February 2026. It functions like a virtual agriculture advisor in multiple Indian languages, offering guidance on crop selection, mandi prices, and eligibility for central schemes such as PM-Kisan, with farmers able to reach an AI assistant by phone.

Q 130. Ans. A

Explanation:

- Supriya Sahu, Additional Chief Secretary (Environment, Climate Change and Forests) of Tamil Nadu, was named a UNEP Champion of the Earth 2025 in the “Inspiration and Action” category. The award, the UN’s highest environmental honour, recognised her work on sustainable cooling (such as the Cool Roof initiative in schools), large-scale tree planting, mangrove restoration, and ecosystem-based climate resilience in Tamil Nadu.
- She was the only Indian among the 2025 laureates.

Q 131. Ans. B

Explanation:

- Malmas Mela (Malmaas Mela) is a major religious fair held in Rajgir, Nalanda district, Bihar, during Adhik Maas (Malmas or Purushottam Maas) of the Hindu lunar calendar. Since Adhik Maas occurs approximately once every three years, the fair is also organised at an interval of about three years and continues throughout the entire sacred month.
- According to local religious belief, 33 crore (33 koti) deities reside in Rajgir during Malmas, making the period exceptionally sacred for pilgrimage and religious observances. The fair attracts thousands of devotees who perform ritual worship and take holy baths in Rajgir’s sacred kunds and hot springs, particularly the Brahmakund, which is believed to cleanse sins, cure ailments, and confer spiritual merit.

Q 132. Ans. C

Explanation:

- **Statement:** Northern Bihar is traversed by several Himalayan rivers, including the **Kosi, Gandak, Bagmati, and Mahananda**, which carry large quantities of sediments eroded from the young Himalayan mountains. During floods and seasonal overflows, these rivers continuously deposit fresh layers of alluvium across the plains of northern Bihar.
- **Conclusion I:** Continuous deposition of fresh alluvium by Himalayan rivers results in the formation of **new alluvial soils (Khadar)** in large parts of northern Bihar. These soils are geologically younger than many older alluvial and residual soils found in parts of southern Bihar. Therefore, this conclusion logically follows from the statement.
- **Conclusion II:** Since fresh sediments are regularly deposited by rivers, the texture, fertility, mineral composition, and other soil properties may vary from

one flood season to another. Thus, soil characteristics in northern Bihar are dynamic and can change over time due to continued alluvial deposition.

● Why Both Conclusions Follow

- ✓ **The statement establishes that northern Bihar receives a continuous supply of fresh sediments from Himalayan rivers. Such deposition:**
 - Produces **young alluvial soils**, supporting Conclusion I.
 - Alters soil composition and properties over time, supporting Conclusion II.

Q 133. Ans. C

Explanation:

- **Statement 1 is correct:** Vidyapati served as a prominent scholar and court poet patronised by the Oiniwar dynasty of Mithila. His most significant royal patron was King Shivasimha, who granted him the village of Bispi and bestowed upon him the title of ‘Abhinava Jayadeva’.
- **Statement 2 is incorrect:** He is historically and culturally celebrated as the “Maithil Kavi Kokil” (the cuckoo poet of Maithili), not the “Magadh Kavi Kokil”. His geographical and linguistic identity is strictly rooted in the Mithila region of northern Bihar, which is distinctly separate from the Magadh region.
- **Statement 3 is correct:** He composed the celebrated *Padavali*, which is widely recognised and compiled in Bengal as the *Bangiya Padabali*. These are lyrical, devotional songs that narrate the sensuous and spiritual love between Radha and Krishna. They became foundational texts for the Vaishnava Bhakti movement in Eastern India.
- **Statement 4 is correct:** Vidyapati’s Maithili poetry gained immense popularity among the medieval poets of Bengal. To emulate his distinct poetic style and rhythm, Bengali Vaishnava poets developed an artificial literary language known as *Brajabuli*, which is essentially a linguistic blend of Maithili and Bengali vocabularies.

Q 134. Ans. B

Explanation:

- The **Bihar Police Chakra App** is an internal digital intelligence and criminal-tracking platform developed by the Bihar Police to assist law enforcement personnel in identifying, monitoring, and analyzing criminal activities in real time. The application enables authorized officers across the state to instantly access criminal records, conduct photograph-based searches, and examine gang

networks and affiliations. It serves as a centralized digital repository containing data on approximately 4.5–5 lakh criminals, compiled over the past decade, thereby enhancing investigative efficiency and inter-district coordination within the police force.

Q 135. Ans. C

Explanation:

- The **Jamui–Jhajha–Chakayi** forest stretch has been recognised as Bihar's first official natural elephant corridor. Located in the forests of Jamui district along the Bihar–Jharkhand border, the corridor serves as an important migratory route for wild elephant herds moving between the two states. It extends across the Garhi Beat of the Jamui Range, the Charkapatthar and Batia Beats of the Jhajha Range, and the Madhwa Sub-Beat of the Chakayi Range. The corridor was officially identified and notified by the Union Ministry of Environment, Forest and Climate Change in its Elephant Corridors of India 2023 report.

Q 136. Ans. A

Explanation:

- **Statement 1 is correct:** According to the **India State of Forest Report (ISFR) 2023**, which includes the Bihar Forest Report 2023, Bihar has a **total forest cover of 7,532 sq km**, constituting approximately **8.0% of the state's geographical area**. The forest cover has increased by **151 sq km** compared to the previous assessment.
 - ✓ Of the total forest cover, **387 sq km** falls under **Very Dense Forest (VDF)**, **3,284 sq km** under **Moderately Dense Forest (MDF)**, and **3,861 sq km** under **Open Forest (OF)**.
 - ✓ The districts with the largest forest cover are **Kaimur (1,025.68 sq km)**, **West Champaran (908.23 sq km)**, **Jamui (670.84 sq km)**, and **Rohtas (668.06 sq km)**.
- **Statement 2 is incorrect:** As per the **ISFR 2023**, Bihar's **tree cover** is **2,371 sq km**, which accounts for approximately **2.5% of the state's geographical area**, not **5.7%**. Tree cover refers to patches of trees occurring outside the recorded forest areas, such as along roads, canals, farms, institutions, and urban spaces. Therefore, the percentage mentioned in the statement is factually incorrect.

Q 137. Ans. B

Explanation:

- **Pair 1 is incorrect:** Shreyasi Singh is a renowned trap shooter from Bihar and one of India's leading

shotgun shooters. She won the **Gold Medal in Women's Double Trap at the 2018 Commonwealth Games (Gold Coast, Australia)** and had earlier secured a **Silver Medal at the 2014 Commonwealth Games (Glasgow)**. She represented India in several international competitions, including the **Paris Olympics 2024**. A native of Jamui district, also serves as the **Sports Minister of Bihar**. She is the daughter of late Digvijay Singh, a former Union Minister and Member of Parliament.

- **Pair 2 is incorrect:** Anshika Kumari is an emerging archer from Bihar who earned selection to the **senior Indian national archery team** for participation in the **Archery World Cup**. She has performed consistently in national-ranking tournaments and age-group competitions, gaining recognition as one of Bihar's promising talents in archery.
- **Pair 3 is incorrect:** Rashmi Kumari is one of India's most accomplished **carrom players** and a multiple-time **World Carrom Champion**. Hailing from Bihar, she has won several **World Championship titles**, numerous **national championships**, and international medals over a career spanning more than two decades.
- **Pair 4 is incorrect:** Sweta Shahi is a pioneering rugby player from Bihar who broke social and sporting barriers to represent India in **Rugby Sevens** at the **Asian Games**. Her participation at the Asian Games made her one of the first women from Bihar to achieve international recognition in rugby.

Q 138. Ans. A

Explanation:

- Thakur Jugal Kishore Sinha (1908–1980) is widely regarded as the **Father of the Cooperative Movement in Bihar**. A distinguished freedom fighter and member of the **First Lok Sabha**, he played a pioneering role in institutionalizing the cooperative sector by advocating the establishment of cooperative departments and strengthening the cooperative credit and agricultural framework across Bihar.
- Janak Singh is recognized as a pioneering figure of the cooperative movement in the **Muzaffarpur and Mithila regions**. Through his leadership and grassroots initiatives, he contributed significantly to the spread of cooperative ideals and organizations among farmers and rural communities in northern Bihar.
- Dr. Verghese Kurien, popularly known as the **Father of the White Revolution in India**, is also remembered in Bihar for his transformative contribution to the state's dairy cooperative sector.

Q 139. Ans. C**Explanation:**

- The Bihar Cabinet has approved the free transfer of 110 acres of land in the Pothia Circle of Kishanganj district for the establishment of a new Central Industrial Security Force (CISF) Regional Training Centre (RTC).
- **Capacity:** The proposed RTC will be developed to train approximately 1,000 CISF personnel simultaneously, significantly enhancing the force's operational preparedness.
- **Strategic Importance:** Kishanganj is located in the Seemanchal region and shares proximity to international borders. It is strategically situated near the Siliguri Corridor ("Chicken's Neck"), a highly sensitive, narrow land bridge connecting mainland India to the Northeastern states.
- **Objective:** The centre is poised to become a vital link in the security strategy for eastern India. It will ensure a permanent presence of the central security apparatus, improve coordination among border agencies (like the SSB and BSF), and generate local employment opportunities in the Seemanchal region.
- During the same period, the Bihar Cabinet also approved the transfer of 100 acres of land in Hajipur to establish the country's third National Institute of Food Technology Entrepreneurship and Management (NIFTEM).

Q 140. Ans. A**Explanation:**

- **Statement:** The service sector contributes the largest share to Bihar's Gross State Domestic Product (GSDP), while a significant proportion of the state's workforce continues to be engaged in agriculture. This indicates that Bihar's economic output is increasingly driven by services, but employment remains concentrated in the primary sector.
- **Conclusion I:** Structural transformation refers to the shift of both **output and employment** from agriculture to industry and services. Although the service sector contributes the largest share to Bihar's GSDP, a majority of workers are still dependent on agriculture. Therefore, the transformation remains **incomplete**, as employment has not shifted proportionately.
- **Conclusion II:** A post-industrial economy is characterised by a dominant service sector in both output and employment, with a relatively small

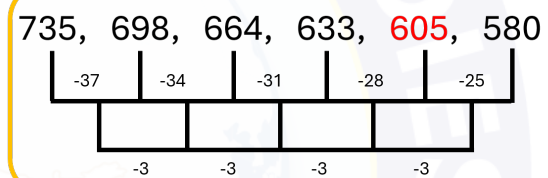
share of workers engaged in agriculture. Since a large proportion of Bihar's workforce is still employed in agriculture, Bihar cannot be described as a post-industrial economy. The statement provides no basis for such a conclusion.

Why Both Conclusions Are Incorrect

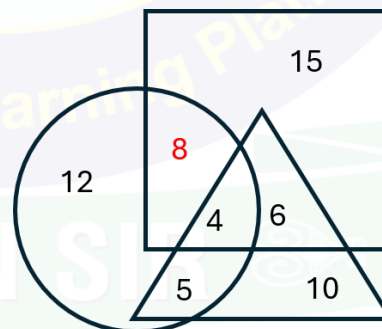
- ✓ The statement highlights a **structural imbalance**: the service sector generates the largest share of income, while agriculture continues to employ most workers. This suggests that Bihar is **undergoing** economic transformation, but neither the structural transformation nor the transition to a post-industrial economy can be considered complete.

Q 141. Ans. A**Explanation:**

Double difference of 3 in consecutive numbers.

**Q 142. Ans. C****Explanation:**

- Circle – Teachers
- Square – Writers
- Triangle – Poets
- Teacher + Writer but not Poet → Circle + Square only
- 8 Teachers are also Writers, but not Poets

**Q 143. Ans. A****Explanation:**

- According to the question,
- Total outcomes = (Total number of balls) = 6 + 5 + 4 = 15
- Favorable outcomes = (Total green balls) = 0
- Required probability = 0/15 = 0

Q 144. Ans. C**Explanation:**

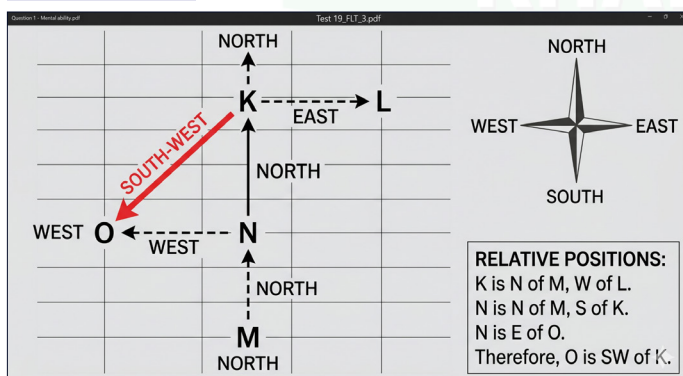
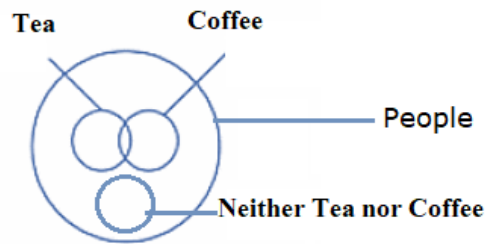
- Ratio of time taken by A and B to complete the race = 200 : 250
- Ratio of their speeds = 250 : 200 = 5 : 4
- Ratio of the distances covered by them = 5 : 4
- $5x = 1000$ meters
- $5x - 4x = x = 200$ meters
- Therefore, A defeated B by 200 meters.

Q 145. Ans. C**Explanation:**

- Number of zeros = Number of factors of 10 ($10 = 2 \times 5$)
- The prime factors of all the numbers are:
 - ✓ $8 = 2^3$
 - ✓ $15 = 3 \times 5$
 - ✓ $25 = 5^2$
 - ✓ $40 = 2^3 \times 5$
 - ✓ $50 = 2 \times 5^2$
 - ✓ $125 = 5^3$
 - ✓ $16 = 2^4$
- Now,
 - ✓ Total power of 2 = $3 + 3 + 1 + 4 = 11$
 - ✓ Total power of 5 = $1 + 2 + 1 + 2 + 3 = 9$
 - ✓ Number of zeros = 9
- Therefore, the number of zeros = 9

Q 146. Ans. B**Explanation:**

- Let the cost price of the item be ₹x.
- According to the question,
 - ✓ $2200 - x = 5(1400 - x)$
 - ✓ $2200 - x = 7000 - 5x$
 - ✓ $4x = 4800$
 - ✓ $x = ₹1200$
- Required selling price = $1200 \times 80/100 = ₹960$
- Therefore, the item should be sold for ₹960 to incur a loss of 20%.

Q 147. Ans. C**Explanation:****Q 148. Ans. B****Explanation:****Q 149. Ans. A****Explanation:**

- To find the actual time from a mirror image:
- Mirror Time + Actual Time = 11:60
- So,
 - ✓ $11:60 - 5:40$
 - ✓ $= 6:20$
- Therefore, the correct answer is 6:20

Q 150. Ans. D**Explanation:**

- Let the principal be P and the annual rate of interest be r%.
- From the simple interest information:
 - ✓ $SI = (P \times r \times 4)/100 = 1600$
- Therefore,
 - ✓ $P \times r = 40000$
- Now use the compound interest information for 2 years:
 - ✓ $CI = P[(1 + r/100)^2 - 1]$
 - $960 = P(2r/100 + r^2/10000)$
- Substituting $P \times r = 40000$:
 - ✓ $960 = (40000/50) + (40000r/10000)$
 - $960 = 800 + 4r$
 - ✓ $4r = 160$
 - ✓ $r = 40$
- Therefore, the annual rate of interest is 40%.

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